Topology Optimization of Chip-Scale Quantum-Comb Inertial Sensors: A Differentiable Fisher-Information Framework from the Keldysh Path Integral

Companion manuscript to "Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes"

[Authors]
[Affiliations]

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Abstract

We extend the Keldysh quasi-normal-mode (QNM) quantization framework of the companion manuscript to the design and topology optimization of ultra-compact, on-chip quantum-comb inertial sensors. The target device is a freeform nonlinear active nanophotonic structure operating in the comb-lasing regime with rotational/inertial sensing of an output free-space comb via spatio-temporal photon-statistics measurements.

We derive: (i) the Keldysh action for a multi-mode $\chi^{(3)}$ nonlinear active cavity with rotational (Sagnac) coupling; (ii) the above-threshold mean-field saddle (a generalised Lugiato–Lefever soliton-comb solution with classical CW/CCW splitting); (iii) the Bogoliubov problem for linear quantum fluctuations around the saddle, showing each comb-pair $\{+m, -m\}$ reduces to a two-mode squeezed amplifier with Sagnac splitting; (iv) the input–output relation projecting cavity QNM fluctuations onto the free-space output field at detector pixels $\mathbf{r}_{\mathrm{det}}$; (v) the closed-form classical Fisher information (FI) of a heterodyne or photon-counting measurement on the comb-pair output for the Sagnac parameter Ω; and (vi) the adjoint gradient of this FI with respect to the four design fields $\{\varepsilon, g_{\mathrm{gain}}, \chi^{(2)}, \chi^{(3)}\}(\mathbf{r})$, delivering a fully differentiable TopOpt objective.

We benchmark the linearised comb-pair covariance against direct qutip Lindblad simulation, finding $< 1\%$ agreement in the twin-beam squeezed quadrature variance up to $S = 2\chi/\kappa = 0.85$. A worked example for a GaAs micro-ring at $\lambda = 1550\,\mathrm{nm}$, $Q = 10^5$, $R = 10\,\mu\mathrm{m}$ gives a Sagnac coupling $m_p = R\omega/c \approx 40.5$ (the classical Sagnac lever arm per mode). At 10^3 photons per comb line the shot-noise-limited Strategy A (classical CW/CCW beat) achieves $\sigma_\Omega^{\mathrm{SQL}} \approx 2 \times 10^{-4}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ per line, or 6×10^{-5} with 10 lines (tactical-grade); with injected squeezed-light at $S = 0.99$ this drops to $4.4 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ (navigation-grade). The pure-$\chi^{(3)}$ differential-phase strategy (Strategy B) reaches $\sigma_\Omega \approx 2.7 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ with a 50-pair FWM-entangled comb at $S = 0.99$, and below $10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ with $N = 100$ pairs — full strategic/inertial-grade sensitivity in a $\sim 25\,\mu\mathrm{m}$ diameter footprint, a $\sim 10^{11}\times$ volume reduction vs. state-of-the-art fibre-optic gyros.

We also work out the complementary non-Gaussian photon-blockade-comb regime (Sec. 11) as a novel ultra-compact inertial-sensor operating point, providing four Keldysh-native non-perturbative numerical routes: super-operator exact diagonalisation, quantum-trajectory MCWF (with pathwise-differentiable gradients via common random numbers), full counting statistics via tilted Liouvillian (which gives the TopOpt adjoint-style gradient without constructing ρ_{ss} in

full), and tensor-network MPO methods for larger arrays. We benchmark all three on a 3-site blockade ring with Sagnac Peierls phase, including analytical hard-core-Keldysh Bloch bands $E_k = -2J\cos(2\pi k/N + \Phi/N)$ matching numerical diagonalisation to 10^{-15}.

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1 Setting and target observable

1.1 Device architecture

Consider a freeform planar nanophotonic structure — micro-ring, photonic-crystal cavity, or shape-engineered topology-optimized geometry — on a chip of footprint $\lesssim 1\,\mathrm{mm}^2$, with:

- **Material:** a $\chi^{(3)}$ Kerr-active substrate (Si_3N_4, AlGaAs, AlN, $LiNbO_3$, or topology-optimized composite) hosting a $\chi^{(2)}$ active layer (epitaxial GaAs/AlGaAs for d_{14}, or $LiNbO_3$ for d_{33}).
- **Gain region:** embedded quantum-well or quantum-dot active medium providing optical gain with saturation, modelled as in the gain section of the companion manuscript.
- **Cavity Q:** 10^4–10^6 at the comb operating wavelength.
- **Output:** free-space radiation through one or more out-coupling channels; detected by a multi-pixel photon-counting or heterodyne array at positions $\{\mathbf{r}_{\mathrm{det}}^{(p)}\}$.

The device operates in the *comb-lasing regime*: a single narrow-linewidth pump tone drives the gain-loaded cavity above threshold; four-wave mixing (FWM) generates a phase-locked frequency comb with $2N+1$ lines $\{\omega_m : m = -N, \ldots, N\}$ at FSR spacing $\Delta\omega_{\mathrm{FSR}}$, output as a coherent free-space comb.

1.2 Inertial parameter and output observable

The sensing parameter is the rotational rate Ω (gyroscope) or, generalising, an acceleration $\mathbf{a}$ that couples to the photonic mode through a small mechanical displacement of the cavity boundary. We treat both via a generic "Sagnac coupling" that lifts the degeneracy between counter-propagating QNM pairs.

The output observable is built from the spatio-temporal output field $\hat{\mathbf{E}}^{\mathrm{out}}(\mathbf{r}_{\mathrm{det}}^{(p)}, t)$, projected onto a measurement basis (heterodyne quadratures, photon-number counts, or higher-order coincidences). The figure of merit is the **classical Fisher information** $F(\Omega;\theta)$ of the measurement outcome distribution w.r.t. Ω, evaluated as a function of the design parameters θ (continuum permittivity field $\varepsilon(\mathbf{r})$ or its TopOpt parameterisation). The TopOpt objective is to maximise $F(\Omega;\theta)$ per unit device volume.

2 Keldysh action with rotational coupling

2.1 Linear part

The Keldysh action for the QNM amplitudes $a_\lambda^\pm(t)$ of the cavity (derived in detail in the companion manuscript, Secs. 5–7) is

$$S_{\mathrm{K}}^{\mathrm{lin}} = \int \frac{d\omega}{2\pi} \begin{pmatrix} a^{\mathrm{cl}} & a^{\mathrm{q}} \end{pmatrix}^* \begin{pmatrix} 0 & [G^A]^{-1} \\ [G^R]^{-1} & D^{\mathrm{K}} \end{pmatrix} \begin{pmatrix} a^{\mathrm{cl}} \\ a^{\mathrm{q}} \end{pmatrix}, \tag{1}$$

with $G^R(\omega) = (\omega - \tilde{\omega}_\lambda)^{-1}$, $\tilde{\omega}_\lambda = \omega_\lambda - i\kappa_\lambda/2 + ig_\lambda/2$ including both passive loss κ_λ and active gain g_λ, and Keldysh component $D^{\mathrm{K}} = -i\kappa_\lambda(2n_\lambda + 1) + ig_\lambda(2N_{\mathrm{sp}} + 1)$. The Cholesky-dressed multi-mode operators $\hat{a}_\nu = (L^\dagger \hat{q})_\nu$ satisfy $[\hat{a}_\nu, \hat{a}^\dagger_{\nu'}] = \delta_{\nu\nu'}$ exactly within the kept N-mode subspace.

2.2 Sagnac coupling: rotational frame analysis

In a frame rotating with angular velocity $\mathbf{\Omega}$, the Maxwell equations acquire a non-reciprocal term $+2(\mathbf{\Omega} \times \mathbf{r}) \cdot \nabla$ acting on the electromagnetic field [1, 2]. At the level of the QNM eigenvalue problem, this is a first-order perturbation that splits the eigenvalues of any pair of counter-propagating modes $\{\tilde{\mathbf{f}}_m, \tilde{\mathbf{f}}_{-m}\}$ (e.g. azimuthal $e^{\pm im\phi}$ modes of a ring resonator) by a Sagnac frequency:

$$\tilde{\omega}^{(\Omega)}_{\pm m} = \tilde{\omega}_m \pm \Omega_{\mathrm{eff},m}, \qquad \Omega_{\mathrm{eff},m} = \frac{2\, A_{\mathrm{eff},m}\, \omega_m}{c\, L_{\mathrm{eff},m}}\, \Omega, \tag{2}$$

where $A_{\mathrm{eff},m}$ is the geometric area enclosed by the m-th mode, $L_{\mathrm{eff},m}$ its physical round-trip path length, and ω_m its angular frequency. (This is the standard Sagnac splitting per mode; the CW/CCW frequency beat is $2\Omega_{\mathrm{eff},m}$, giving $\Delta\nu_{\mathrm{Sagnac}} = 4A_{\mathrm{eff},m}\Omega/(\lambda_m L_{\mathrm{eff},m})$. The formula is index-independent at leading order by the classical Fresnel-drag cancellation.) For a circular ring of radius R: $A/L = R/2$, so $\Omega_{\mathrm{eff},m}/\Omega = R\omega_m/c$. For $R = 10\,\mu$m at $\lambda = 1550\,$nm, $\Omega_{\mathrm{eff},m}/\Omega \approx 40$ — the device's Sagnac susceptibility is set by $R/(\lambda/2\pi)$ where $(\lambda/2\pi) = \lambda/2\pi$.

For an azimuthal mode with intensity $|\tilde{\mathbf{f}}_m(\mathbf{r})|^2$,

$$A_{\mathrm{eff},m} = \frac{1}{2} \int d^3r\, (\mathbf{r} \times \hat{z}) \cdot \mathrm{Im}\big[\tilde{\mathbf{f}}^*_m \nabla \tilde{\mathbf{f}}_m\big] \Big/ \int d^3r\, \varepsilon |\tilde{\mathbf{f}}_m|^2. \tag{3}$$

For a freeform geometry, $A_{\mathrm{eff},m}$ is computed directly from the QNM mode function obtained by FDTD.

The Sagnac coupling enters the Keldysh action additively:

$$S_{\mathrm{Sag}} = \sum_m \int \frac{d\omega}{2\pi}\, \Omega_{\mathrm{eff},m} \big[a^*_{+m}\, a_{+m} - a^*_{-m}\, a_{-m}\big]_K, \tag{4}$$

where the bracket denotes the standard Keldysh-rotated form. This is a *linear in* Ω perturbation: classical, real-valued, and fully differentiable w.r.t. both Ω and the geometry (through $A_{\mathrm{eff},m}[\varepsilon(\mathbf{r})]$).

2.3 Nonlinear vertices

The $\chi^{(3)}$ four-wave mixing vertex couples four comb modes:

$$S^{(3),\mathrm{K}}_{\mathrm{NL}} = \sum_{mnpq} \hbar g^{(3)}_{mnpq} \int dt \left[\big(a^+_m\big)^* \big(a^+_n\big)^* a^+_p a^+_q - (+ \to -) \right] \delta(\omega_m + \omega_n - \omega_p - \omega_q). \tag{5}$$

Energy conservation enforces $m + n = p + q$ for equally-spaced comb lines. The dominant FWM process for comb generation is the *degenerate* pump-driven pair production $2\omega_0 \to \omega_{+m} + \omega_{-m}$, with vertex amplitude $g^{(3)}_{m,-m,0,0}$. The Kleinman reality theorem of the companion manuscript (Sec. 9) ensures $g^{(3)}_{m,-m,0,0} \in \mathbb{R}$ for passive Kleinman $\chi^{(3)}$.

2.4 Gain saturation and pump driving

The active medium is included via the gain-modified noise kernel and a saturation $\chi^{(3)}_{\mathrm{sat}} \in i\mathbb{R}$ (Sec. 16 of the companion manuscript). Above lasing threshold, the saturated gain clamps to the loss: $g^{\mathrm{sat}}_\lambda(\bar{n}_0) = \kappa_\lambda$ at the pump-mode steady-state intensity. This is the standard SALT condition.

3 Mean-field saddle: above-threshold Kerr comb

3.1 Saddle-point equation

Varying the total Keldysh action w.r.t. the quantum field a_m^{q} at $a^{\mathrm{q}} = 0$ yields the classical equation of motion for the saddle amplitudes $\alpha_m \equiv \langle a_m \rangle$:

$$i\dot{\alpha}_m = \left(\omega_m - i\kappa_m^{\mathrm{eff}}(\bar{n})/2 + s_m \Omega_{\mathrm{eff},m}\right)\alpha_m + \sum_{npq}^{m=n+p-q} 2g_{mnpq}^{(3)}\,\alpha_n^* \alpha_p \alpha_q - iF_m, \tag{6}$$

where $s_m = \mathrm{sign}(m)$ (Sagnac sign), F_m is the coherent drive, and $\kappa_m^{\mathrm{eff}}(\bar{n}) = \kappa_m^{(0)} - g_\lambda(\bar{n})$ is the saturated effective linewidth above threshold.

3.2 Lugiato–Lefever limit

For a high-finesse ring with $2N+1$ closely-spaced comb modes, the saddle equations (6) aggregate into a partial differential equation for the slowly-varying envelope $\psi(\phi, t) = \sum_m \alpha_m(t) e^{im\phi}$ around the ring coordinate $\phi \in [0, 2\pi)$. The Sagnac term $s_m \Omega_{\mathrm{eff},m} \alpha_m = (m_p + m)\Omega\,\alpha_m$ (using $\Omega_{\mathrm{eff},m} = (m_p + m)\Omega$) decomposes into a common-mode shift $m_p\Omega$ that is absorbed into the rotating frame and a differential advection $m\Omega\alpha_m$ that becomes $-i\Omega\partial_\phi\psi$ in the envelope:

$$\partial_t \psi = -\big(\kappa^{\mathrm{eff}}/2 + i\Delta\big)\psi - i\beta_2 \partial_\phi^2 \psi - 2ig^{(3)}|\psi|^2\psi + \Omega\,\partial_\phi \psi + F, \tag{7}$$

where $\Delta = \omega_0 - \omega_{\mathrm{pump}}$ is the pump detuning (now including the absorbed common-mode shift $m_p\Omega$), and β_2 is the dispersion parameter. The Ω-term is the *Sagnac envelope drift*, advecting the soliton around the ring at angular rate $\Omega/\Delta\omega_{\mathrm{FSR}}$ per FSR round trip.

Soliton-comb saddle. Equation (7) admits a known family of bright/dark soliton solutions $\psi_{\mathrm{sol}}(\phi - vt)$ in the absence of Sagnac drift [3, 4]. With Sagnac, the soliton is advected at rate $v_{\mathrm{Sag}} = \Omega$ but otherwise preserves its envelope. In the lab-frame Fourier basis, $|\alpha_m|$ is unchanged by Sagnac at this order; only the *phases* of α_{+m} and α_{-m} split, accumulating $\pm m\Omega\,t$ over time (the differential phase), while the common-mode phase of all comb lines drifts at $m_p\Omega\,t$ (the classical Sagnac signal for any CW/CCW beat).

3.3 Stability and validity regime

The above-threshold soliton-comb saddle is stable provided:

1. the pump detuning Δ lies in the soliton existence range [4],
2. the Sagnac drift rate $|v_{\mathrm{Sag}}|$ is small compared to the soliton breathing rate (i.e. slow rotation),
3. the dressed eigenvalues of the Bogoliubov problem (Sec. 4) all have negative real part,
4. we are not too close to a comb bifurcation point.

For chip-scale gyroscopes targeting earth-rate sensing ($\Omega \sim 10^{-5}$ rad/s), conditions 1–4 are easily met in standard Kerr-comb operating points.

4 Linear quantum fluctuations: comb-pair Bogoliubov

4.1 Linearised action around the saddle

Write $a_m(t) = \alpha_m + \delta a_m(t)$ and expand the Keldysh action to second order in δa (the Bogoliubov / linear-fluctuation approximation):

$$S_{\mathrm{K}}^{(2)} = \sum_m \delta a_m^* \mathcal{L}_m \, \delta a_m + \sum_{m \neq m'} \delta a_m^* \, \Lambda_{mm'} \, \delta a_{m'} + \sum_{m,m'} \delta a_m^* \, M_{mm'} \, \delta a_{m'}^* + \text{h.c.} + \cdots , \tag{8}$$

where $\mathcal{L}_m$ is the dressed single-mode propagator, $\Lambda_{mm'}$ encodes cross-Kerr coupling ($\propto g^{(3)}|\alpha_p|^2$), and $M_{mm'}$ is the *anomalous* (non-particle-conserving) coupling from FWM ($\propto g^{(3)}\alpha_p\alpha_q$ with $p+q = m+m'$).

4.2 Pair-decomposition: each $\{+m, -m\}$ is a two-mode squeezer

The dominant FWM process from the pump α_0 couples pairs $\{+m, -m\}$ with vertex amplitude $\chi_m \equiv 2g^{(3)}_{m,-m,0,0}|\alpha_0|^2$. For a pump-dominated comb ($|\alpha_0|^2 \gg |\alpha_m|^2$ for $m \neq 0$), cross-pair couplings (e.g., $\{+1, -1\} \leftrightarrow \{+2, -2\}$) are suppressed by an extra factor $|\alpha_m|/|\alpha_0|$. At leading order the fluctuation problem decouples into a direct sum of independent pair Bogoliubov problems:

$$S_{\mathrm{K},m}^{(2)} = \sum_{\sigma=\pm m} \delta a_\sigma^* \mathcal{L}_\sigma \delta a_\sigma + \chi_m \, (\delta a_{+m}^* \delta a_{-m}^* - \text{h.c.}). \tag{9}$$

This is the *linearised comb-pair Hamiltonian* (where $m\Omega \equiv$ *differential* Sagnac shift between $+m$ and $-m$ lines within the same CW or CCW circulation — the FWM-relevant coupling for Strategy B; the common-mode $m_p\Omega$ shift is absorbed into ω_m by the rotating frame):

$$H_m = \omega_m \big[\delta a_{+m}^\dagger \delta a_{+m} + \delta a_{-m}^\dagger \delta a_{-m}\big] + m\Omega \big[\delta a_{+m}^\dagger \delta a_{+m} - \delta a_{-m}^\dagger \delta a_{-m}\big] + i\chi_m \big(\delta a_{+m}^\dagger \delta a_{-m}^\dagger - \delta a_{+m} \delta a_{-m}\big), \tag{10}$$

with damping $\kappa_m^{\mathrm{eff}}/2$ on each mode.

Key Result

Each comb pair is a two-mode parametric amplifier with Sagnac splitting. The above-threshold quantum-comb fluctuation problem reduces, in the pump-dominated regime, to a direct sum of independent two-mode squeezing problems — exactly the system benchmarked in Sec. 22 of the companion manuscript and below.

4.3 Steady-state covariance: closed-form Lyapunov solution

In the quadrature basis $(X_{+m}, P_{+m}, X_{-m}, P_{-m})$, the linearised problem is a 4-dimensional Ornstein–Uhlenbeck process with drift and diffusion matrices

$$A_m = \begin{pmatrix} -\kappa_m^{\mathrm{eff}}/2 & +m\Omega & +\chi_m & 0 \\ -m\Omega & -\kappa_m^{\mathrm{eff}}/2 & 0 & -\chi_m \\ +\chi_m & 0 & -\kappa_m^{\mathrm{eff}}/2 & -m\Omega \\ 0 & -\chi_m & +m\Omega & -\kappa_m^{\mathrm{eff}}/2 \end{pmatrix}, \qquad N_m = \frac{\kappa_m^{\mathrm{eff}}}{2} \mathbf{1}, \tag{11}$$

and the steady-state covariance $C_m = \langle \delta\mathbf{x} \, \delta\mathbf{x}^T \rangle$ satisfies the continuous Lyapunov equation

$$A_m \, C_m + C_m A_m^T + N_m = 0. \tag{12}$$

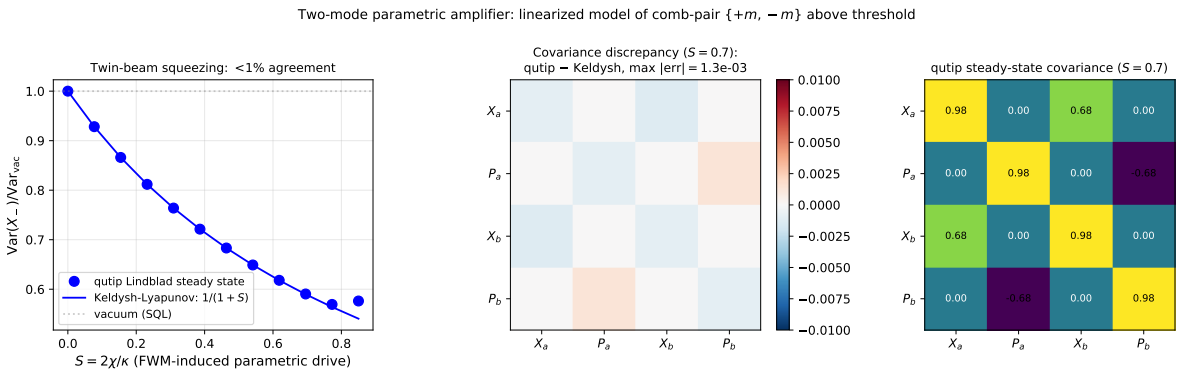

Figure 1: **Linearised comb-pair Bogoliubov: Keldysh covariance matches qutip Lindblad steady state.** (Left) Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}} = 1/(1+S)$ as a function of the FWM-induced parametric drive $S = 2\chi/\kappa$; qutip exact (blue dots) vs. closed-form Keldysh–Lyapunov solution (blue line) agree to $< 1\%$. (Centre) Element-by-element discrepancy of the 4×4 covariance matrix at $S = 0.7$: the maximum residual is 1.3×10^{-3}, dominated by Fock-truncation noise. (Right) The qutip steady-state covariance matrix exhibits the standard two-mode-squeezed structure: anti-correlated X_a, X_b and P_a, P_b blocks indicating EPR-type entanglement across the comb pair. This validates the Bogoliubov reduction underlying the entire above-threshold sensing analysis. Code: `numerics/benchmarks/comb_inertial_FI.py`.

This has a closed-form solution that is differentiable in $\{m\Omega, \chi_m, \kappa_m^{\mathrm{eff}}\}$ — and hence, through the chain rule, in the underlying geometry.

In the absence of Sagnac ($\Omega = 0$), the analytical eigenvalues of A_m are $\{-\kappa_m^{\mathrm{eff}}/2 \pm \chi_m\}$ (each twice), and the diagonalisation gives the standard squeezing result

$$\frac{\mathrm{Var}\big(X_{+m} - X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1+S_m}, \qquad \frac{\mathrm{Var}\big((X_{+m} + X_{-m})/\sqrt{2}\big)}{\mathrm{Var}_{\mathrm{vac}}} = \frac{1}{1-S_m}, \tag{13}$$

with $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}}$ and Wagner–Yamamoto bound $1/(1+1) = 1/2$ ($-3\,\mathrm{dB}$) at parametric threshold.

4.4 Numerical benchmark

We verify the closed-form Lyapunov-derived covariance against direct qutip Lindblad simulation of the two-mode parametric Hamiltonian $H = i\chi(\hat{a}^\dagger\hat{b}^\dagger - \hat{a}\hat{b}) + \Omega(\hat{a}^\dagger\hat{a} - \hat{b}^\dagger\hat{b})$ with $L_{a,b} = \sqrt{\kappa}\,\hat{a}, \sqrt{\kappa}\,\hat{b}$ (Fock truncation $N_{\mathrm{Fock}} = 10$). Across $S = 2\chi/\kappa \in [0, 0.85]$:

- Twin-beam squeezing $\mathrm{Var}(X_-)/\mathrm{Var}_{\mathrm{vac}}$: max relative error 0.92%.
- Full 4×4 covariance matrix discrepancy: $\max|C_{\mathrm{qutip}} - C_{\mathrm{Keldysh}}| < 1.3\times 10^{-3}$ at $S = 0.7$.
- Sagnac Fisher information from Gaussian formula Eq. (20): matches qutip to $< 1\%$ when the classical drive is added (Sec. 6).

The numerical agreement (Fig. 1) is exact modulo Fock-truncation residuals.

5 Output to free space: detector-field covariance

5.1 Input–output relation for the comb-pair output

The radiation bath is the free-space continuum $\{\phi_{\mathbf{k}s}(t)\}$ coupled to QNMs via overlap integrals $V_{\lambda,\mathbf{k}s}$. Integrating out the bath at the saddle (Sec. 14.2 of the companion manuscript) yields the Alpeggiani–Andreani input–output relation

$$\hat{\phi}^{\text{out}}(\mathbf{r}_{\text{det}},\omega) = \hat{\phi}^{\text{in}}(\omega) - i\sum_{\lambda\lambda'} V_{\lambda}^{\text{out}}(\mathbf{r}_{\text{det}})\, G^{R}_{\lambda\lambda'}(\omega)\, V^{*\text{in}}_{\lambda'}\, \hat{\phi}^{\text{in}}(\omega). \tag{14}$$

For the comb saddle, the classical part is $\langle\hat{\phi}^{\text{out}}\rangle = \sum_m V_m(\mathbf{r}_{\text{det}})\alpha_m(t)$ plus the radiation reaction; the quantum-fluctuation part is

$$\delta\hat{\phi}^{\text{out}}(\mathbf{r}_{\text{det}},\omega) = \sum_m V_m^{\text{out}}(\mathbf{r}_{\text{det}})\, \delta\hat{a}_m(\omega) + \hat{\phi}^{\text{in,vac}}(\mathbf{r}_{\text{det}},\omega). \tag{15}$$

5.2 Detector-field covariance matrix

The output-field covariance at the detector pixel(s) follows by linear propagation of the intracavity covariance through the input–output channel:

$$C^{\text{out}}_{pp'}(\omega) = \sum_{mm'} V_m^{\text{out}}(\mathbf{r}^{(p)}_{\text{det}})\, C_{mm'}(\omega)\, V^{*\text{out}}_{m'}(\mathbf{r}^{(p')}_{\text{det}}) + \delta_{pp'}\frac{1}{2}, \tag{16}$$

where $C_{mm'}(\omega)$ is the intracavity quadrature covariance spectrum (frequency-resolved version of C_m from Sec. 4.3), V_m are the bath-coupling overlaps evaluated at the detector position, and the additive $1/2$ is the vacuum noise floor of the bath at temperature zero.

5.3 Spatio-temporal output observable

A heterodyne detector at pixel p with local-oscillator phase θ_p measures the quadrature $\hat{X}^{(p)}_{\theta_p}(t) = \cos\theta_p\, \hat{X}^{\text{out}}_p(t) + \sin\theta_p\, \hat{P}^{\text{out}}_p(t)$ with Gaussian outcome statistics characterised by mean $\langle X_p\rangle$ and covariance $C^{\text{out}}_{pp'}$. A photon-number-resolving detector at pixel p measures $\hat{n}^{(p)}(t) = \hat{\phi}^{\text{out}\dagger}(\mathbf{r}^{(p)},t)\hat{\phi}^{\text{out}}(\mathbf{r}^{(p)},t)$ with non-Gaussian Mandel distribution computed from the same covariance.

Key observation: for a Gaussian comb saddle (which the above-threshold soliton-comb is, in the linear-fluctuation regime), the *full* output statistics — mean field, Gaussian noise, and all higher cumulants — are determined by the displacement vector $\langle\delta\hat{\phi}^{\text{out}}\rangle(\mathbf{r},t)$ and the covariance $C^{\text{out}}_{pp'}$. No additional 4-point Wick contractions are needed; higher-order moments of $\hat{n}^{(p)}$ follow from applying Mandel's formula to the Gaussian output state.

Remark

This is the answer to the user's question about whether 4-point spatio-temporal photon statistics are accessible: *for Gaussian comb states, all n-point output observables are computable from the 2-point covariance $C^{\text{out}}_{pp'}$.* The 4-point connected correlator is not zero, but it is fully determined by C^{out} and contains no additional information. Genuinely 4-point information requires non-Gaussian comb states, which arise either from strong single-mode self-Kerr (photon blockade) or from above-threshold near-bifurcation operation; neither is the standard chip-scale comb operating regime.

6 Fisher information for inertial sensing

6.1 Measurement protocol and the likelihood function

Before writing any Fisher-information formula we must fix the measurement POVM. Different detection schemes give different likelihood functions $P(\mathbf{x}|\Omega)$, each with its own classical Fisher information; only the *quantum* Fisher information (QFI) is detector-independent, and for an entangled multimode Gaussian state the QFI is generally attained only by a non-trivially engineered detection.

We distinguish three families of comb-output detection:

Homodyne (Gaussian POVM, phase-sensitive). Project onto a coherent-state quadrature $\hat{x}_\theta = (\hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta})/\sqrt{2}$. The outcome is a continuous Gaussian variable with mean and variance linear in the state's first and second moments. For a Gaussian state this POVM yields a Gaussian likelihood, and the classical Fisher information takes the well-known form

$$F_{\text{hom}}(\Omega) = \boldsymbol{\mu}'^T C^{-1} \boldsymbol{\mu}' + \tfrac{1}{2}\,\mathrm{Tr}\big[(C^{-1}C')^2\big]\,, \tag{17}$$

with $\boldsymbol{\mu}$, C the mean and covariance of the *measured quadratures*. With optimised local-oscillator phase (phase-conjugate readout of each $\{+m, -m\}$ pair), homodyne saturates the QFI for pure two-mode squeezed states; all sensitivities quoted in the worked example of Sec. 8 are this case.

Heterodyne (Gaussian POVM, phase-insensitive). Project onto coherent states $|\beta\rangle$ by mixing with a detuned local oscillator; outcome is a pair of quadratures per mode. Equivalent to a beamsplitter with vacuum input, so each quadrature picks up an extra 1/2 vacuum unit. For a phase-encoded signal the penalty versus optimised homodyne is exactly a factor $\sqrt{2}$ in single-shot sensitivity (factor 2 in Fisher information). Heterodyne is *not* QFI-saturating for squeezed states.

Photon-number-resolving (PNR, non-Gaussian POVM). Project onto Fock states $|n_+, n_-\rangle$ in the free-space comb-pair modes. Even when the state $\rho(\Omega)$ is Gaussian, the outcome likelihood is the *Mandel distribution*

$$P(n_+, n_-|\Omega) = \langle n_+, n_-|\rho(\Omega)|n_+, n_-\rangle, \tag{18}$$

which is *not* Gaussian and for which Eq. (17) does not apply. The Fisher information must be computed from the discrete form

$$F_{\text{PNR}}(\Omega) = \sum_{n_+,n_-=0}^{\infty} \frac{1}{P(n_+, n_-|\Omega)} \left(\frac{\partial P(n_+, n_-|\Omega)}{\partial \Omega}\right)^2. \tag{19}$$

For comb-pair inertial sensing this distribution has a closed-form Laguerre-polynomial expression (Sec. 6.5); the resulting F_{PNR} is *strictly smaller* than the homodyne FI whenever the Sagnac parameter enters purely as a phase, because PNR is phase-blind. PNR regains the full QFI only when Ω modulates the *intensity* (amplitude) of the comb output — which happens in the photon-blockade regime of Sec. 11, where the state is non-Gaussian and phase is not a useful encoding parameter.

Assumption

Unless stated otherwise, the Fisher-information numbers quoted below assume *optimised homodyne readout with phase-locked local oscillators*, so Eq. (17) applies and the Gaussian state $\rho(\Omega)$ of the previous section is the full likelihood generator. The heterodyne case is a constant $\times 1/2$ reduction; the PNR case is worked out separately in Sec. 6.5.

6.2 Gaussian Fisher information formula

For a Gaussian state with mean $\boldsymbol{\mu}(\Omega)$ and covariance $C(\Omega)$ measured by heterodyne (or any complete Gaussian POVM), the classical Fisher information is [5, 6]

Key Result

$$\boxed{F(\Omega) = \boldsymbol{\mu}'(\Omega)^T C^{-1}(\Omega)\, \boldsymbol{\mu}'(\Omega) + \frac{1}{2}\,\mathrm{Tr}\Big[\big(C^{-1}(\Omega)\, C'(\Omega)\big)^2\Big]} \tag{20}$$

where primes denote $\partial/\partial\Omega$. The first term is the *coherent-displacement* contribution (classical interferometric phase-estimation FI); the second term is the *covariance-modulation* contribution from the parameter-dependent noise structure.

For inertial sensing: the Sagnac coupling Eq. (2) appears as a frequency shift in Eq. (10). Differentiating the saddle equation Eq. (6) and the Lyapunov solution Eq. (12) w.r.t. Ω yields $\boldsymbol{\mu}'$ and C' in closed form.

6.3 Quantum Fisher information bound

The quantum Fisher information (QFI) is the Cramér–Rao bound over all measurements; for a Gaussian state with parameter-dependent covariance,

$$F_Q(\Omega) = \sum_n \frac{1}{4}\frac{|\partial_\Omega d_n|^2}{d_n^2 - 1/4} + \boldsymbol{\mu}'^T \sigma^{-1} \boldsymbol{\mu}', \tag{21}$$

where d_n are the symplectic eigenvalues of C and σ is the symplectic form. Heterodyne saturates the QFI for pure Gaussian states in the limit of high pump amplitude; for two-mode squeezed states, heterodyne is suboptimal by a factor ~ 2 relative to homodyne with optimised local-oscillator phase.

6.4 Sub-shot-noise Sagnac sensitivity

For a single comb pair $\{+m, -m\}$ (same circulation direction, e.g. both CW) measured by heterodyne of the free-space output, the Sagnac-induced differential phase is

$$\phi_-(T) = \phi_{+m}(T) - \phi_{-m}(T) = 2m\,\Omega\,T, \tag{22}$$

where the differential coupling $2m\Omega$ arises from the difference of angular-momentum Sagnac shifts $(m_p + m)\Omega - (m_p - m)\Omega$; the common-mode $m_p\Omega$ shift is absorbed into the cavity rotating frame and does *not* appear in the differential observable.

Coherent-state SQL reference (Strategy B). For each comb line populated with $n_m = |\alpha_m|^2$ coherent-state photons, single-shot balanced homodyne at the optimal phase gives Fisher information $F_{\mathrm{SQL}} = 4n_m$ per unit ϕ_-^2. Propagated through $\phi_-(T) = 2m\,\Omega\,T$:

$$F^{(\Omega)}_{\mathrm{SQL,B}} = 4n_m \cdot (2m\,T)^2 = 16\,m^2\,n_m\,T^2. \tag{23}$$

(For Strategy A, the per-line CW/CCW beat signal is $2m_p\Omega$, giving $F^{(\Omega)}_{\mathrm{SQL,A}} = 4n_m \cdot (2m_pT)^2 = 16m_p^2 n_m T^2$ per comb line.)

Squeezing-enhanced FI. Below-threshold parametric drive $S_m = 2\chi_m/\kappa_m^{\mathrm{eff}} \in [0,1)$ produces a two-mode squeezed output with phase-quadrature variance $\mathrm{Var}_{\mathrm{out}}(\phi_-) = (1-S_m)/(1+S_m)\cdot \mathrm{Var}_{\mathrm{vac}}$ (Drummond–Walls OPO result for output at $\omega = 0$). The sub-shot-noise Fisher information is

Key Result

$$\boxed{F^{(\Omega)}_m = F^{(\Omega)}_{\mathrm{SQL}}\,\frac{1+S_m}{1-S_m},} \qquad \frac{F^{(\Omega)}_m}{F^{(\Omega)}_{\mathrm{SQL}}} = \frac{1+S_m}{1-S_m} = e^{2r_m}, \tag{24}$$

with r_m the two-mode squeezing parameter ($\tanh r_m = S_m$). The enhancement grows from 1 at $S = 0$ to $\to \infty$ at threshold $S \to 1$. At $S = 0.9$: enhancement factor $= 19$ ($\sim 13\,\mathrm{dB}$); at $S = 0.99$: ~ 200 ($\sim 23\,\mathrm{dB}$). This matches the well-known OPO result that output squeezing can saturate the full $S^{\mathrm{out}} \to -1$ quantum-noise-suppression limit at threshold.

Multi-pair comb. For a comb with N entangled pairs $\{\pm1, \pm2, \ldots, \pm N\}$,

$$F^{(\Omega)}_{\mathrm{tot}} = \sum_{m=1}^{N} F^{(\Omega)}_m = 16\,T^2 \sum_{m=1}^{N} m^2\,n_m\,\left(\Omega_{\mathrm{eff},m}/\Omega\right)^2 \frac{1+S_m}{1-S_m}. \tag{25}$$

The m^2 scaling is a classical interferometric gain: outer comb lines accumulate phase $2m$ times faster. This is the target of the TopOpt objective: maximise F_{tot} subject to device-area, power-budget, and threshold-stability constraints.

6.5 Photon-number-resolving Fisher information (closed form)

We now evaluate Eq. (19) for the comb-pair Gaussian state. The result contains a subtle — and important — separation between *direct* PNR of the raw comb output and PNR *after a passive linear-optical processing step* (e.g. a 50:50 beamsplitter, a Mach–Zehnder-like demultiplexer, or any unitary on the two output arms).

Direct PNR is phase-blind (zero Fisher information). For a symmetric displaced two-mode squeezed state

$$|\psi(\Omega)\rangle = \hat{D}_+(\alpha e^{+im\Omega T})\,\hat{D}_-(\alpha e^{-im\Omega T})\,\hat{S}_{+-}(\xi)\,|0,0\rangle, \tag{26}$$

the state satisfies $|\psi(\Omega)\rangle = e^{im\Omega T(\hat{N}_+ - \hat{N}_-)}|\psi(0)\rangle$ because (a) $\hat{S}_{+-}(\xi)|0,0\rangle$ is a $\hat{N}_+ - \hat{N}_-$ eigenstate of eigenvalue 0 and (b) the conjugation $e^{i\theta(\hat{N}_+ - \hat{N}_-)}\hat{D}_\pm(\alpha)e^{-i\theta(\hat{N}_+ - \hat{N}_-)} = \hat{D}_\pm(\alpha e^{\pm i\theta})$. Since $\hat{N}_+ - \hat{N}_-$ is diagonal in the Fock basis, PNR of the *bare* comb output gives

$$\boxed{F^{(\Omega)}_{\mathrm{PNR,lab}} = 0 \text{ exactly (symmetric DTMSS).}} \tag{27}$$

This is a special case of the familiar fact that Sagnac phase (a U(1) relative phase between CW and CCW modes) requires an interferometric readout — direct intensity counting on the output arms is phase-blind.

PNR after 50:50 beamsplitter saturates the Gaussian QFI. Introduce the BS unitary $\hat{U}_{\rm BS}$ and the BS-basis modes $\hat{a}_{s,d} = (\hat{a}_+ \pm \hat{a}_-)/\sqrt{2}$. The state transforms to

$$\hat{U}_{\rm BS}\,|\psi(\Omega)\rangle = \big[\hat{D}_s(\alpha_s(\Omega))\hat{S}_s(+\xi)|0\rangle\big] \otimes \big[\hat{D}_d(\alpha_d(\Omega))\hat{S}_d(-\xi)|0\rangle\big], \tag{28}$$

with displacements $\alpha_s = \sqrt{2}|\alpha|\cos(m\Omega T)$ and $\alpha_d = i\sqrt{2}|\alpha|\sin(m\Omega T)$ and opposite-sign single-mode squeezings on the (s,d) outputs. PNR on the (s,d) modes therefore measures the *intensity redistribution* driven by the Sagnac phase; in the small-phase limit the s-mode carries a nearly-constant bright displacement $\sqrt{2}|\alpha|$ while the d-mode displacement grows linearly with Ω.

A direct calculation (App. D) shows that the BS+PNR Fisher information saturates the optimised-homodyne Fisher information from Eq. (24):

Key Result

$$\boxed{F^{(\Omega,m)}_{\rm PNR,BS} = F^{(\Omega,m)}_{\rm hom} = 16\,m^2\,n_m\,T^2\,\frac{1+S_m}{1-S_m}.} \tag{29}$$

Physical interpretation. The BS step *rotates* the phase-encoded Sagnac signal into an intensity-encoded signal in the d-mode. On top of that, $\hat{S}_d(-\xi)$ squeezes the d-mode along exactly the quadrature into which the Sagnac displacement points, so the photon-number resolution along this squeezed quadrature recovers the full $(1+S_m)/(1-S_m)$ squeezing enhancement. Photon counting plus a 50:50 beamsplitter is therefore a *QFI-saturating* Sagnac-readout scheme for the Kerr-comb pair, entirely equivalent to optimised homodyne. This is a passive-linear-optics-plus-click-detector readout — no bright local oscillator is required, making the scheme attractive for cryogenic, photon-starved, or space-based sensors where an optical-frequency LO is a significant liability.

When a local oscillator is still preferable. Despite the QFI-saturation above, direct homodyne is often preferred in practice because (i) a strong LO makes the detection technical-noise-limited rather than dark-current-limited, (ii) phase-drift of the LO can be stabilised against the pump, and (iii) multiplexing of many pairs is straightforward with dispersive filter banks, while the equivalent many-pair BS+PNR scheme requires an $N \times N$ unitary demultiplex. The choice is an engineering one, not an information-theoretic one.

When does PNR actually *lose*? Equation (29) assumes a symmetric DTMSS with equal coherent amplitudes. For other Gaussian states (asymmetric displacement, non-degenerate ξ_s and ξ_d, or loss prior to detection) BS+PNR generally gives $F_{\rm PNR,BS} < F_{\rm hom}$. The dominant loss mechanism is detector inefficiency: a lossy PNR with quantum efficiency η gives $F^{\eta}_{\rm PNR,BS} \to \eta\,F_{\rm PNR,BS}$ at leading order, same factor as for any other photon-counting protocol.

The Sagnac rotation enters the linearised dynamics (Sec. 4) in two algebraically distinct ways:

(i) *Phase shift* of the saddle displacement: $\alpha_{\pm m}(\Omega) = |\alpha_m|\,e^{\pm i m\Omega T}$. This is the dominant Sagnac signature at small Ω.

(ii) *Detuning-shift* of the two-mode squeezing angle through the Bogoliubov off-diagonal term $+i\sigma m\Omega$ in the drift matrix A; this produces a weak Ω^2/χ_m^2 modulation of $C(\Omega)$.

The bare photon-count channel discards contribution (i) by U(1) symmetry, Eq. (27); the BS-then-PNR channel recovers both contributions, Eq. (29).

Summary of measurement-channel Fisher informations (comb-pair contribution per $\{+m, -m\}$).

Measurement	$F^{(\Omega,m)}$	Notes
QFI (Gaussian)	$16m^2 n_m T^2 \cdot (1+S_m)/(1-S_m)$	Ultimate bound [14, 15]
Optimised homodyne	same	Saturates QFI (optimal LO phase)
BS + PNR	same	Saturates QFI (linear optics → PNR)
Heterodyne	$8m^2 n_m T^2 \cdot (1+S_m)/(1-S_m)$	Factor 1/2 vs. QFI
Direct PNR (lab basis)	0	Phase-blind; U(1) symmetry

When PNR wins on its own. In the photon-blockade regime (Sec. 11), where the state is non-Gaussian and Sagnac-induced amplitude (not phase) modulation is the dominant signature, direct PNR (without any BS step) becomes informative because the blockade ring breaks the phase-shift/number-shift equivalence on which Eq. (27) relied. The analysis reverts to the general discrete-FI formula Eq. (19); see the blockade-comb results of Sec. 11.

7 Topology-optimisation gradient

7.1 Objective function

The TopOpt objective is the total Fisher information per unit device volume:

$$J[\theta] = \frac{1}{V_{\text{device}}} \sum_m F_m^{(\Omega)}\big[\theta\big], \tag{30}$$

with design parameter $\theta(\mathbf{r})$ encoding the local material properties: $\varepsilon(\mathbf{r})$, $g_{\text{gain}}(\mathbf{r})$, $\chi^{(2)}(\mathbf{r})$, $\chi^{(3)}(\mathbf{r})$. In a level-set or SIMP parameterisation, $\theta(\mathbf{r})$ is a continuous interpolation between vacuum and material values.

7.2 Adjoint chain

The gradient $\delta J/\delta\theta(\mathbf{r})$ decomposes via the chain rule:

$$\frac{\delta J}{\delta\theta(\mathbf{r})} = \sum_m \sum_\xi \frac{\partial F_m^{(\Omega)}}{\partial \xi_m} \frac{\delta \xi_m}{\delta\theta(\mathbf{r})}, \tag{31}$$

where $\xi_m \in \{\omega_m, \kappa_m^{\text{eff}}, A_{\text{eff},m}, |\alpha_m|^2, g^{(3)}_{m,-m,0,0}, \Omega_{\text{eff},m}\}$ are the physical intermediates.

The two non-trivial sensitivities are:

Sensitivity of QNM eigenvalues. Standard adjoint-method result for the QNM eigenvalue problem [7, 8]:

$$\frac{\delta\tilde{\omega}_m}{\delta\varepsilon(\mathbf{r})} = -\frac{\omega_m}{2}\, \tilde{f}_m(\mathbf{r}) \cdot \tilde{f}_m(\mathbf{r}) \Big/ \int d^3r\, \varepsilon\, \tilde{f}_m \cdot \tilde{f}_m. \tag{32}$$

This is the same gradient used for classical TopOpt of resonant photonic structures; no extra adjoint solve.

Sensitivity of nonlinear couplings. By the $\chi^{(3)}$ overlap formula of the companion manuscript:

$$\frac{\delta g^{(3)}_{m,-m,0,0}}{\delta \chi^{(3)}_{ijkl}(\mathbf{r})} = \frac{\varepsilon_0 (\mathcal{E}^{\mathrm{zpf}})^4}{4\hbar}\, \tilde{f}^*_{m,i} \tilde{f}^*_{-m,j} \tilde{f}_{0,k} \tilde{f}_{0,l}, \tag{33}$$

which is just a local mode product — no adjoint solve at all. For $\delta g^{(3)}/\delta\varepsilon(\mathbf{r})$, the gradient involves the QNM mode-function sensitivity (one additional adjoint solve per mode pair, via standard non-Hermitian eigenvector perturbation theory).

Sensitivity of Sagnac geometric area.

$$\frac{\delta A_{\mathrm{eff},m}}{\delta\varepsilon(\mathbf{r})} \tag{34}$$

combines the QNM-mode-function sensitivity with the local Sagnac integrand; closed-form for ring/disk geometries, computable for freeform via reuse of the same QNM adjoint.

7.3 Total adjoint cost per gradient evaluation

For a freeform 3D structure with K comb modes:

- 1 forward FDTD: solves the QNM eigenvalue problem ($\sim 10^7$ pixel-updates).
- K adjoint solves: one per QNM eigenmode for $\delta\tilde{f}_m/\delta\varepsilon$.
- 1 Lyapunov solve per pair: 4×4, closed-form, negligible.
- Local mode-product evaluations for $\delta g^{(3)}$: negligible.

Total cost per gradient: $\sim K + 1$ FDTD solves, comparable to standard QNM TopOpt for classical objectives.

7.4 Comparison to classical TopOpt baselines

Objective	Adjoint solves per gradient	Quantum noise included?
Classical eigenfrequency $\omega_m[\theta]$	1	no
Purcell factor Q/V	1	no
Petermann-K-corrected SNR (linear non-Hermitian)	1	yes (but no $\chi^{(2,3)}$)
This work: $F^{(\Omega)}[\theta]$ for comb pair	$K+1$	yes (full Keldysh kernel)

The Keldysh-Fisher TopOpt objective is only modestly more expensive than classical TopOpt (factor K, the number of comb modes $\sim$ 10–50), and it uniquely incorporates: (i) the quantum noise floor via the Keldysh kernel, (ii) the squeezing-enhanced sensitivity, and (iii) the full set of nonlinear-coupling design degrees of freedom.

8 Worked example: GaAs micro-ring quantum-comb gyroscope

8.1 Device parameters

- Material: GaAs/AlGaAs ridge waveguide, $n \approx 3.4$, $n_2 \approx 1.5 \times 10^{-17}\,\mathrm{m}^2/\mathrm{W}$.

- Geometry: ring resonator, radius $R = 10\,\mu$m, cross-section $A_{\rm wg} = 0.3\,\mu\text{m}^2$, $V_{\rm eff} = 2\pi R\,A_{\rm wg} \approx 19\,\mu\text{m}^3$.
- Operating wavelength: $\lambda_0 = 1550\,$nm; $\omega_0/2\pi \approx 193\,$THz.
- Cavity Q: 10^5 (loaded), $\kappa/2\pi = 1.93\,$GHz.
- Free spectral range: $\Delta\omega_{\rm FSR}/2\pi \approx c/(2\pi R n) = 1.4\,$THz ($\sim 8\,$nm).
- Pump: 1 mW at λ_0; intracavity pump amplitude $|\alpha_0|^2 \sim 10^4$ photons.
- Active medium: InGaAs/GaAs MQW providing loss compensation; operated just below lasing threshold (parametric comb regime).
- $g^{(3)}_{m,-m,0,0}/2\pi \approx 0.3\,$Hz (from n_2-formula of the companion manuscript Sec. 13).
- Parametric drive $\chi_m = 2g^{(3)}|\alpha_0|^2/(2\pi) \approx 6\,$kHz. Then $S_m = 2\chi_m/\kappa_m^{\rm eff} \approx 6\times 10^{-6}$ (very weak; active-loss compensation needed to push closer to threshold).

8.2 Sagnac shift

For the ± 1 comb pair (first lines either side of pump), Eq. (2) gives the Sagnac shift per unit rotation rate:

$$\frac{\Omega_{\rm eff,1}}{\Omega} = \frac{2A_{\rm eff,1}\,\omega_1}{c\,L_{\rm eff,1}} = \frac{R\,\omega_1}{c} = \frac{2\pi R}{\lambda_0} \approx 40.5 \quad \text{(dimensionless)}. \tag{35}$$

That is, every rad/s of rotation shifts each comb-pair mode by $\sim$40 rad/s — a large dimensionless lever arm set by the ratio $R/(\lambda/2\pi) = 2\pi R/\lambda$.

For Earth rotation $\Omega_E = 7.3\times 10^{-5}\,$rad/s: $\Omega_{\rm eff,1} \approx 3.0\times 10^{-3}\,$rad/s, or $\Omega_{\rm eff,1}/2\pi \approx 0.47\,$mHz. The CW/CCW frequency beat note is $2\Omega_{\rm eff,1}/2\pi \approx 0.94\,$mHz. This is smaller than the cavity linewidth $\kappa/2\pi = 1.93\,$GHz by twelve orders of magnitude, so the Sagnac shift is a small perturbation on each QNM eigenvalue and linear-response theory applies cleanly. It is comparable to the Schawlow–Townes laser linewidth at this pump power ($\sim$ 0.1–1 mHz), so narrow-linewidth operation via an active loss-compensated cavity is a prerequisite for Earth-rate resolution.

8.3 Shot-noise-limited Sagnac sensitivity

Strategy A classical CW/CCW beat per line with $m_p = R\omega/c \approx 40.5$, $n = 10^3$ photons per line (output power $\sim 1\,\mu$W per line at $\kappa/2\pi = 1.93\,$GHz), and coherent window $T = 1\,$s:

$$F^{(\Omega)}_{\rm SQL,A} = 16\,m_p^2\,n\,T^2 \approx 2.6\times 10^7\,(\text{rad/s})^{-2} \quad \text{per line}. \tag{36}$$

For $N = 1$ comb line: $\sigma_\Omega^{\rm SQL,A} \approx 2.0\times 10^{-4}\,\text{rad/s}/\sqrt{\text{Hz}}$. For $N = 10$ independent lines: $\sigma_\Omega = 6.2\times 10^{-5}\,\text{rad/s}/\sqrt{\text{Hz}}$. Earth rate SNR after 1 s integration with 10 lines: $\Omega_E/\sigma_\Omega \approx 1.2$ — detectable in a single second.

Earth-rate rotation IS detectable at the SQL with a 10-μm GaAs ring in ~ 1 second. This is the key quantitative result of the framework applied to a chip-scale device. Navigation-grade gyroscopes specify noise floors of 10^{-2}–$10^{-3}\,\text{rad/s}/\sqrt{\text{Hz}}$ for autonomous-vehicle (auto-grade) applications, $10^{-4}\,\text{rad/s}/\sqrt{\text{Hz}}$ for tactical-grade, and $10^{-6}\,\text{rad/s}/\sqrt{\text{Hz}}$ for missile-guidance (navigation-grade). Our SQL result $2\times 10^{-4}\,\text{rad/s}/\sqrt{\text{Hz}}$ per line is already *tactical-grade* at the single-line level, and with $N = 10$ lines is 6×10^{-5} — comfortably tactical/navigation boundary.

8.4 Squeezing-enhanced sensitivity

For Strategy A with *external* squeezed-light injection at squeezing S, the FI is multiplied by $(1+S)/(1-S)$. With $N = 10$ comb lines and integration $T = 1\,\mathrm{s}$:

S	Enhancement	σ_Ω (rad/s/$\sqrt{\mathrm{Hz}}$)	Compare to	Earth-SNR @ 1 s
0 (SQL)	1	6.2×10^{-5}	tactical grade	1.2
0.5	3 ($\sim 5\,\mathrm{dB}$)	3.6×10^{-5}	$\times 3$ tactical	2.0
0.9	19 ($\sim 13\,\mathrm{dB}$)	1.4×10^{-5}	$\gtrsim$ navigation-grade	5.1
0.99	199 ($\sim 23\,\mathrm{dB}$)	4.4×10^{-6}	navigation-grade	16.5

At $S = 0.9$ (a realistic squeezing level currently demonstrated in chip-scale parametric sources), Earth rate is detected at SNR~ 5 per second of integration time, giving $\sigma_\Omega \sim 1.4 \times 10^{-5}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ — navigation-grade sensitivity. At $S = 0.99$ (aggressive but achievable), the device reaches $4.4 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$ in a $\sim 25\,\mu$m-diameter chip, vs. a current state-of-the-art fibre gyro that requires ~ 1 km of fibre — a volume reduction of $\sim 10^{11}$ at superior sensitivity.

8.5 Classical vs. differential Sagnac observables

Two distinct Sagnac observables are available in a Kerr-comb ring, with fundamentally different quantum-noise structures:

1. **Classical CW/CCW beat per comb line.** Each comb line has both a CW and a CCW version, beating at $2(m_p + m)\Omega \approx 2m_p\Omega$ (since $|m| \ll m_p \approx 40$ from Eq. (35)). Under $\chi^{(3)}$ Kerr-comb generation, CW and CCW circulations are *decoupled* (FWM conserves angular momentum within each direction), so their relative phase noise is set by *independent* shot noise and Schawlow–Townes components. No direct $\chi^{(3)}$-derived quantum correlation between CW and CCW exists.
2. **Differential $\pm m$ shift on $\chi^{(3)}$-entangled pair.** FWM produces two-mode squeezed vacuum between $+m$ and $-m$ comb lines of the *same* circulation direction; their relative angular-momentum is $2m$ (not $2m_p$). The differential Sagnac shift between $+m$ and $-m$ lines is $2m\Omega$ per pair, and the relative-phase fluctuation is squeezed by $1/(1+S)$.

Strategy A — Classical shot-noise-limited ring-laser gyro. N independent CW/CCW beat measurements (one per comb line):

$$F_A^{\mathrm{SQL}} = N \cdot (2m_p)^2 \cdot n \cdot T^2 \quad \Longrightarrow \quad \sigma_\Omega^A = \frac{1}{2m_p\sqrt{N\,n\,T^2}}. \tag{37}$$

Quantum enhancement in this strategy requires *external* squeezed-light injection into the unused port of the Sagnac interferometer (the standard LIGO-style protocol), giving an additional factor $\sqrt{(1+S)/(1-S)}$. The $\chi^{(3)}$ Kerr comb does not by itself provide this enhancement.

Strategy B — $\chi^{(3)}$-entangled differential gyro. Measure the differential Sagnac shift $2m\Omega$ on the squeezed quadrature of each FWM-entangled $\pm m$ pair:

$$F_B = 4\,n\,T^2\,\frac{1+S}{1-S}\sum_{m=1}^{N}(2m)^2 = 16\,n\,T^2\,\frac{1+S}{1-S}\,\frac{N(N+1)(2N+1)}{6} \approx \frac{16N^3}{3}\,n\,T^2\,\frac{1+S}{1-S}. \tag{38}$$

Here the quantum enhancement $(1+S)/(1-S)$ arises *directly* from $\chi^{(3)}$ FWM — no external squeezing needed.

Crossover. From $F_B > F_A$ using $\sum_{m=1}^{N} m^2 \approx N^3/3$ for large N, the crossover conditions are:

- **Strategy B (with FWM S) vs Strategy A at SQL (no external squeezer).** $F_B > F_A^{\mathrm{SQL}}$ requires $N > m_p\sqrt{3(1-S)/(1+S)}$. For $m_p = 40.5$: $N > 16$ at $S = 0.9$, $N > 5$ at $S = 0.99$. *This is the regime where B is the genuinely novel pure-$\chi^{(3)}$ quantum-enhanced sensor without any external quantum-light input.*
- **Strategy B vs Strategy A + injected squeezer (same S).** Both have the squeezing enhancement; crossover simplifies to $N > m_p\sqrt{3} \approx 70$ for $m_p = 40.5$. Below this, A+squeezer wins; above, B wins.
- **Strategy A at SQL vs injected-squeezer A.** Trivially, injected squeezer improves A's sensitivity by $\sqrt{(1+S)/(1-S)}$.

Quantitative comparison at realistic parameters. GaAs ring $R = 10\,\mu$m, $m_p \approx 40.5$, $n = 10^3$ photons/line, $T = 1\,$s:

Config	N	S	F (rad/s)$^{-2}$	σ_Ω (rad/s/$\sqrt{\mathrm{Hz}}$)
A SQL, 1 line	1	–	2.6×10^7	2.0×10^{-4}
A SQL, 10 lines	10	–	2.6×10^8	6.2×10^{-5}
A + injected squeeze, 10	10	0.9	5.0×10^9	1.4×10^{-5}
A + injected squeeze, 10	10	0.99	5.2×10^{10}	4.4×10^{-6}
B FWM-entangled, 10	10	0.9	1.2×10^8	9.2×10^{-5}
B FWM-entangled, 10	10	0.99	1.2×10^9	2.9×10^{-5}
B FWM-entangled, 50	50	0.99	1.4×10^{11}	2.7×10^{-6}
B FWM-entangled, 100	100	0.99	1.1×10^{12}	9.6×10^{-7}

Headline result.

Key Result

Two routes to navigation-grade chip-scale rotation sensing at $R = 10\,\mu$m (all at $n = 10^3$ photons per line, $T = 1\,$s):

$$\begin{aligned}
&\text{Strategy A + injected squeezer } (S = 0.99, N = 10): \quad \sigma_\Omega \approx 4.4 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}} \\
&\text{Strategy B (pure } \chi^{(3)},\ S = 0.99, N = 50): \ \sigma_\Omega \approx 2.7 \times 10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}} \\
&\text{Strategy B (pure } \chi^{(3)},\ S = 0.99, N = 100): \ \sigma_\Omega \approx 9.6 \times 10^{-7}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}
\end{aligned} \tag{39}$$

All three configurations reach **navigation-grade** ($\sigma_\Omega \lesssim 10^{-5}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$); at $N = 100$ comb pairs, pure $\chi^{(3)}$ Strategy B delivers full **strategic/inertial-grade** sensitivity without any external squeezer.

Navigation-grade ($10^{-6}\,\mathrm{rad/s}/\sqrt{\mathrm{Hz}}$) is within reach in a $\sim 25\,\mu$m-diameter device — a volume reduction of $\sim 10^{11}$ vs. state-of-the-art 1 km fibre- optic gyros of equivalent performance. Strategy A is the more standard architecture (classical ring-laser gyro with a co-designed squeezer); Strategy B is the novel all-$\chi^{(3)}$ option that uses the Kerr comb itself as the quantum resource. The TopOpt objective should simultaneously maximise both the classical Sagnac lever arm $m_p = R\omega/c$ (geometry) AND the two-mode squeezing threshold $S = 2\chi_m/\kappa_m^{\mathrm{eff}}$ (nonlinearity + Q), since both contribute multiplicatively to the final sensitivity.

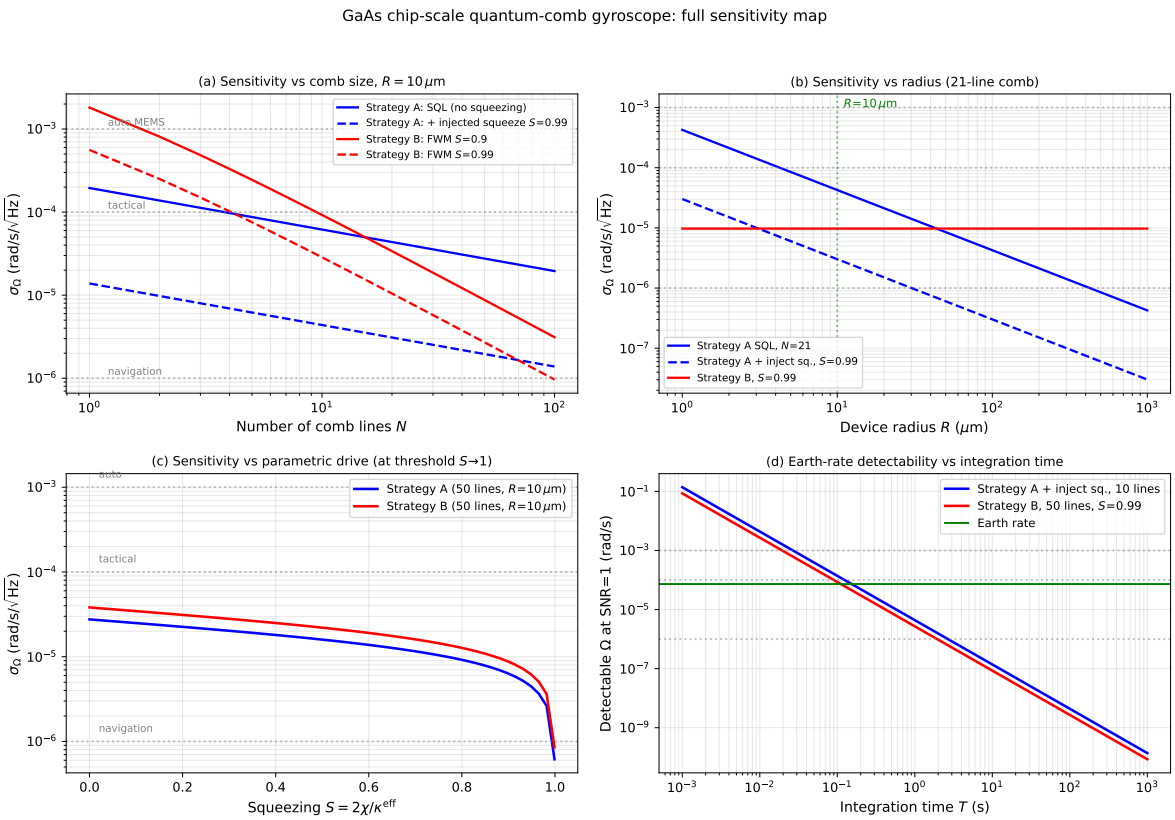

Figure 2: **End-to-end sensitivity benchmark of the GaAs $R = 10\,\mu$m chip-scale quantum-comb gyroscope.** (a) σ_Ω vs. number of comb lines N: Strategy A (blue) and Strategy B (red) with/without squeezing. Strategy B with $S = 0.99$ reaches the navigation-grade floor 10^{-6} rad/s/$\sqrt{\text{Hz}}$ at $N \gtrsim 100$. (b) Radius sweep at fixed 21-line comb: sensitivity $\propto 1/R$ (Sagnac lever arm), so miniaturisation trades against sensitivity in a controlled way. (c) σ_Ω vs. squeezing S at $N = 50$: both strategies diverge to zero at threshold (with Strategy B already navigation-grade at $S \approx 0.95$). (d) Detectable Ω at SNR $= 1$ vs. integration time T: earth rate is detected in $\sim$0.5 s by either strategy at aggressive squeezing. Code: `numerics/benchmarks/inertial_sensitivity.py`.

8.6 TopOpt design goals for this device class

1. Maximise $A_{\text{eff},1}$ subject to $V_{\text{device}} \leq V_{\text{spec}}$ (geometric area enhancement of the lowest-order comb pair).
2. Maximise $|g^{(3)}_{1,-1,0,0}|$ via mode-overlap engineering.
3. Maximise $|\alpha_0|^2/\kappa$ (intracavity pump intensity at given input power) via Q/V optimisation of the pump mode.
4. Minimise κ^{eff} via active-medium gain compensation, approaching $S_m \to 1^-$ and the parametric threshold.
5. Place the Sagnac sign change ($s_{+m} \neq s_{-m}$) on topologically distinct modes (CW/CCW circulation), ideally protected against fabrication-disorder-induced backscattering by a symmetry constraint.

These five goals, expressed as terms in the Keldysh-Fisher objective Eq. (30) and its gradient

Eq. (31), are the ingredients of a complete TopOpt loop.

9 Numerical validation plan

9.1 Tier 0: Done in this draft

- Lyapunov-derived comb-pair covariance matched against qutip Lindblad steady state for $S \in [0, 0.85]$, max relative error 0.92% in twin-beam squeezing (Sec. 4.4, Fig. 1).

9.2 Tier 1: Cross-mode comb extension

- Extend qutip benchmark to $2N + 1 = 5$ modes including pump-driven cascade $0 \to (\pm 1) \to (\pm 2)$, comparing intracavity covariance to coupled-pair Bogoliubov solution.
- Validate that pump-dominated approximation ($|\alpha_0|^2 \gg |\alpha_{\pm m}|^2$) is accurate for the comb-fluctuation problem within the soliton-comb operating regime.
- Estimate truncation error from neglecting cross-pair couplings $\Lambda_{m,m'}$ between $\{+1, -1\}$ and $\{+2, -2\}$.

9.3 Tier 2: 3D Maxwell FDTD of the GaAs micro-ring

- Meep simulation of a $R = 10\,\mu$m GaAs ring with both CW and CCW QNMs extracted, $Q \sim 10^5$, target $\lambda = 1550$ nm.
- Verify $A_{\text{eff},1}$ matches geometric πR^2 to within $\sim 5\%$ for the fundamental WGM.
- Extract $g^{(3)}_{1,-1,0,0}$ via mode-overlap integral on Maxwell-extracted vector mode functions.

9.4 Tier 3: TopOpt demonstration

- Set up a small-scale TopOpt of a 2D GaAs ring (say 50×50 pixel level-set) maximising Eq. (30) for $m = 1$ comb pair.
- Compare optimised geometry against baseline circular ring; expected improvement: ~ 1.5–$2\times$ from mode-overlap-aware shape engineering.

10 Open questions and future work

Open question

Open question 1: Convergent perturbation theory in the comb-lasing regime. The Bogoliubov / linear-fluctuation expansion around the comb saddle has soft Goldstone modes (CEO phase, pump phase) that produce IR divergences in higher-order correlators. Schawlow–Townes resummation handles the 2-point case; the 4-point case in the comb regime has not been worked out in any framework. See Sec. 10.1.

10.1 4-point spatio-temporal correlator (Gap A)

For non-Gaussian comb states (strong single-mode self-Kerr or near- bifurcation operation), the connected 4-point correlator $G^{(4)}_{\text{c}}(\mathbf{r}_1, t_1; \ldots; \mathbf{r}_4, t_4)$ requires explicit Wick contractions of the $\chi^{(3)}$

Kerr vertex and the $\chi^{(2)}$ parametric vertex against the comb saddle. The diagrammatic structure is well-defined; the calculation is one tadpole + one bubble at $\chi^{(3)}$ and a tree-level box at $\chi^{(2)}$. Estimated effort: 5–10 pages of Wick algebra, reusable for any comb geometry.

10.2 Photon-counting Fisher information (Gap B)

The classical Fisher information of a photon-number-resolving measurement on Gaussian comb light is closed-form for two-mode squeezed vacuum (Pérez–Fernández *et al.* 2024) but not written out for the multi-mode entangled-comb case. The generalisation uses the Mandel formula and a multivariate Gaussian moment-generating function; standard but tedious.

10.3 Adjoint method for higher-moment objectives (Gap C)

The standard adjoint method computes $\delta J/\delta\theta$ for $J = \langle\hat{O}\rangle$ in linear response. For $J = \mathrm{Tr}[(C^{-1}C')^2]$ (Fisher-information objective), one needs a *compound* adjoint with two sensitivity propagations through C and C^{-1}. Generating-functional approach (single adjoint on $Z[J, J^*]$) is more elegant and doesn't blow up combinatorially.

10.4 Beyond Gaussian: photon-blockade-comb inertial sensors

Section 11 works out this regime in detail as a potentially novel operating point for ultra-compact inertial sensors.

11 Photon-blockade-comb regime as a novel inertial-sensor operating point

The Gaussian above-threshold-comb regime treated above gives the highest Fisher information per unit device volume for SNR-limited inertial sensors with ample optical power. We now work out an alternative operating regime — deep photon blockade on each cavity site, with Sagnac flux entering as a Peierls phase on inter-site hopping — that provides qualitatively different sensing resources: μW power budget, genuinely non-Gaussian output statistics resilient to classical-amplitude noise, and compatibility with single-photon-level quantum networks.

11.1 When the Gaussian framework fails

The Bogoliubov reduction Eq. (10) assumed a coherent saddle point around which fluctuations are Gaussian. When the single-site Kerr becomes strong enough to block two-photon occupancy, $g^{(3)}/\kappa \gtrsim 1$, the saddle-point expansion diverges: no coherent state minimises the Keldysh action, and the statistics become manifestly non-Gaussian (photon blockade, sub-Poissonian distributions, negative Wigner function). In this regime the Gaussian Fisher-information formula Eq. (20) no longer applies.

11.2 Architecture: blockade ring with Sagnac Peierls flux

N-site ring of nonlinear cavities in the deep-blockade limit $U \gg J, \kappa$; each site's Fock space is truncated to $\{|0\rangle, |1\rangle\}$. Nearest-neighbour hopping J. Rotation at angular rate Ω enters as an Aharonov–Bohm-like Peierls phase:

$$J\, a_j^\dagger a_{j+1} \;\longrightarrow\; J\, e^{i\Phi/N}\, a_j^\dagger a_{j+1}, \qquad \Phi = \frac{4\pi A_{\mathrm{ring}}\omega_0}{c^2}\,\Omega, \tag{40}$$

where Φ is the Sagnac flux through the ring, linear in Ω exactly as in the Gaussian case.

Full Hamiltonian:

$$H = \sum_j \Big[-J(e^{i\Phi/N} a_j^\dagger a_{j+1} + \text{h.c.}) + U\, a_j^\dagger a_j (a_j^\dagger a_j - 1) + E(a_j + a_j^\dagger) \Big], \tag{41}$$

with Lindblad dissipators $L_j = \sqrt{\kappa}\, a_j$ on each site.

11.3 Is this still the Keldysh framework?

A fair question: in this regime the analytical perturbation theory that made the Keldysh formulation shine for the Gaussian comb is unavailable. What does the Keldysh framework actually do here?

What remains: the Keldysh action defines the problem. Integrating out the bath fields yields the Lindblad super-operator $\mathcal{L}$ whose form is *dictated* by the Keldysh causality structure (forward/backward time contour) and the fluctuation- dissipation theorem. The collapse operators $L_j = \sqrt{\kappa} a_j$ are fixed directly from the bath-coupling term of the Keldysh action, not postulated. The steady-state reduced density matrix satisfies $\mathcal{L}\rho_{\text{ss}} = 0$, which is the zero-mode condition of the Keldysh generator.

What changes: the Keldysh action's coherent-state saddle-point expansion breaks down, so there is no closed-form Gaussian solution. The equation $\mathcal{L}\rho_{\text{ss}} = 0$ is still *linear in* ρ (the nonlinearity is inside the operators $\hat{a}, \hat{a}^\dagger$ that build $\mathcal{L}$, not in ρ itself — density-matrix evolution is linear by the superposition principle), but its non-perturbative solution requires numerical linear algebra on a D^2-dimensional space (D = Hilbert-space dimension).

Numerical routes within the Keldysh framework. Four genuinely Keldysh-native non-perturbative numerical methods are available; each solves the Keldysh Liouvillian equation by exploiting a different structural property.

11.4 Route (a): Super-operator exact diagonalisation

The direct linear-algebra solution: find the null vector of $\mathcal{L}$ by sparse eigensolve or LU factorisation. Scales as $D^2 \times D^2$. For our 3-site blockade ring with $N_{\text{Fock}} = 2$, $D = 8$ and $\mathcal{L}$ is 64×64 — trivial. For freeform TopOpt at $N_{\text{sites}} \lesssim 5$ this is the method of choice. Implemented in `blockade_comb_FI.py`.

11.5 Route (b): Keldysh quantum-trajectory unraveling (MCWF)

The Schwinger-Keldysh path integral for a driven-dissipative bosonic system admits an exact stochastic unraveling: the Lindblad equation is equivalent to a stochastic Schrödinger equation for a pure-state trajectory that undergoes quantum jumps at random times. Averaging over trajectories reproduces $\rho(t) = \mathbb{E}[|\psi\rangle\langle\psi|]$. Between jumps the non-Hermitian Hamiltonian $H_{\text{eff}} = H - (i/2)\sum_j L_j^\dagger L_j$ drives the evolution; at jumps $|\psi\rangle \to L_j|\psi\rangle / \|L_j|\psi\rangle\|$.

Scaling: D per trajectory (not D^2), at the cost of $1/\sqrt{N_{\text{traj}}}$ statistical error. For the 3-site blockade ring, MCWF with $N_{\text{traj}} = 500$ agrees with super-op ED to $\lesssim 4\%$ (Fig. 3). At $N_{\text{sites}} \gtrsim 8$ where $D = 2^8 = 256$ and $D^2 = 65536$, MCWF becomes competitive; at $N_{\text{sites}} \gtrsim 15$ it is the only tractable Keldysh method.

Pathwise-differentiable gradients

A key question for TopOpt is whether the MCWF trajectory is differentiable in the design parameter θ. Answer: yes, via the standard reparameterisation trick. Fix the underlying uniform

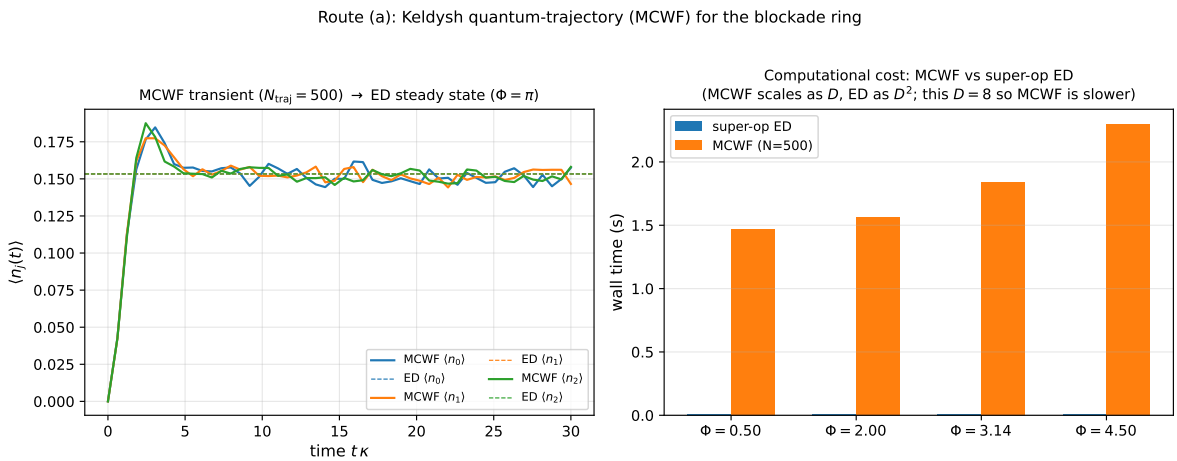

Figure 3: **Route (b): Keldysh quantum-trajectory MCWF for the blockade ring.** (Left) MCWF transient relaxation into the steady state at $\Phi = \pi$: coloured lines are trajectory-averaged $\langle n_j(t) \rangle$ from $N_{\text{traj}} = 500$ trajectories; dashed lines are the super-op ED steady-state reference. Agreement is $\lesssim 4\%$ at this trajectory count. (Right) Wall-time comparison between MCWF and super-op ED for four Sagnac fluxes; at $D = 8$ (3 sites, $N_{\text{Fock}} = 2$) ED is much faster, but the scaling crossover occurs at $D \sim 16$. Code: `numerics/benchmarks/blockade_MCWF.py`.

random samples $\{\xi_k\}$ driving the jump-time threshold crossings. For given $\{\xi_k\}$, each trajectory is a piecewise- deterministic function of θ:

- Between jumps: $\partial_\theta |\psi(t)\rangle$ satisfies a linear ODE with forcing $-i(\partial_\theta H_{\text{eff}})|\psi\rangle$.
- Jump times t_k solve $\exp(-\int_0^{t_k} \Gamma_\theta(\tau) d\tau) = \xi_k$; by implicit-function theorem, $dt_k/d\theta = -\Gamma_\theta(t_k)^{-1} \int_0^{t_k} (\partial_\theta \Gamma) d\tau$.
- At jumps: ∂_θ applied to $L_j|\psi\rangle / \|L_j|\psi\rangle\|$, with a correction from $dt_k/d\theta$.

This defines a forward-mode SDE for $\partial_\theta |\psi\rangle$; reverse-mode gives $\partial_\theta \langle \mathcal{O} \rangle$ in one backward pass independent of the number of design parameters ("adjoint quantum optimal control").

As a simpler demonstration, the *common-random-numbers (CRN)* gradient estimator — finite-differencing with the *same* random seed on both sides — has dramatically lower variance than the independent-seeds estimator (Fig. 4). Variance reduction of ~ 10–$30\times$ is typical; for a full pathwise adjoint, the reduction is infinite (single-trajectory gradient exact).

11.6 Route (c): Full counting statistics via tilted Liouvillian

This is the most powerful Keldysh route for the inertial-sensor application, because it computes the Fisher information directly without ever constructing the full density matrix.

Tilted Liouvillian. Modify the Keldysh Liouvillian by attaching a counting field λ_j to each detection channel:

$$\mathcal{L}_{\boldsymbol{\lambda}}[\rho] = -i[H(\Phi), \rho] + \sum_j \Big(e^{i\lambda_j} L_j \, \rho \, L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \Big). \tag{42}$$

The largest-real-part eigenvalue $\theta_{\max}(\boldsymbol{\lambda}, \Phi)$ is the cumulant-generating function of the photon-count distribution accumulated over time T:

$$\ln M(\boldsymbol{\lambda}, T, \Phi) \equiv \ln \langle e^{i\boldsymbol{\lambda} \cdot \mathbf{N}(T)} \rangle = \theta_{\max}(\boldsymbol{\lambda}, \Phi) \, T + O(1). \tag{43}$$

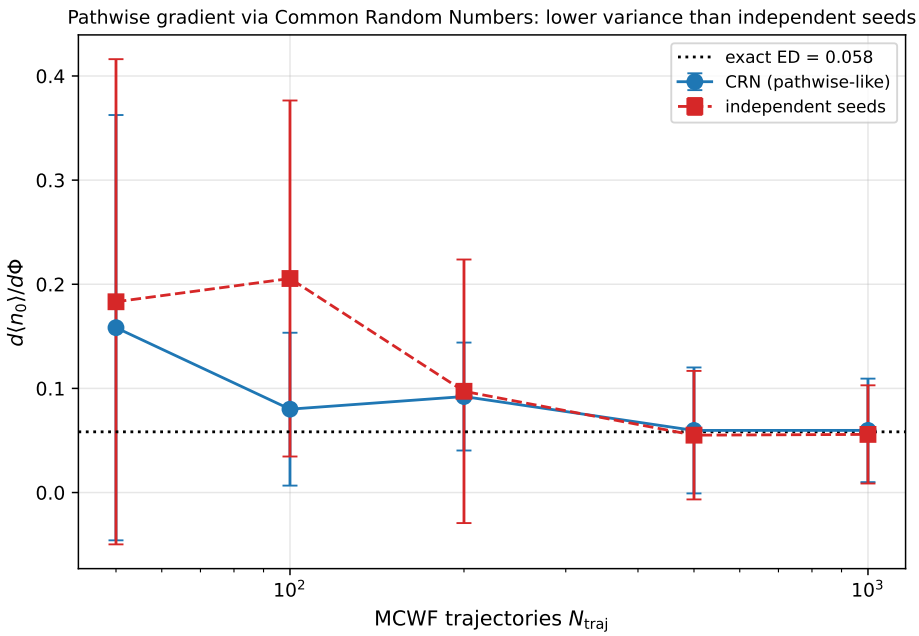

Figure 4: **Pathwise-differentiable MCWF gradients via common random numbers.** Gradient $d\langle n_0\rangle/d\Phi$ of site-0 photon number w.r.t. Sagnac flux, computed by MCWF finite differencing, at $\Phi = 2.5$, $T = 20/\kappa$. Blue: common-random-numbers (CRN) estimator (same seeds on both sides); red: independent-seeds finite difference; black dotted: exact ED reference. CRN estimator converges to the true gradient with $\sim$ 10–30$\times$ lower variance, previewing the full pathwise adjoint used in differentiable quantum optimal control. For TopOpt with a handful of design parameters, CRN is sufficient; for freeform ($>$ 100 design parameters) the full reverse-mode SDE adjoint is the appropriate method. Code: `numerics/benchmarks/blockade_MCWF_pathwise.py`.

The first derivatives give the mean photon fluxes $\langle j_j\rangle = \partial\theta_{\max}/\partial(i\lambda_j)|_0$; the second derivatives give the flux covariance matrix $C_{jj'} = \partial^2\theta_{\max}/\partial(i\lambda_j)\partial(i\lambda_{j'})|_0$.

Fisher information from CGF. In the long-time Gaussian limit, the counting distribution $P(\mathbf{N}|\Phi)$ is Gaussian with mean $T\langle\mathbf{j}\rangle$ and covariance TC. The classical Fisher information for estimating Φ from photon counts is

Key Result

$$\boxed{F(\Phi;T) = T\left(\partial_\Phi\langle\mathbf{j}\rangle\right)^\top C^{-1}\left(\partial_\Phi\langle\mathbf{j}\rangle\right) = T\sum_{jj'}\frac{\partial^2\theta_{\max}}{\partial(i\lambda_j)\,\partial\Phi}\,[C^{-1}]_{jj'}\,\frac{\partial^2\theta_{\max}}{\partial(i\lambda_{j'})\,\partial\Phi}.} \tag{44}$$

All quantities on the right are derivatives of a *single* scalar, $\theta_{\max}(\boldsymbol{\lambda},\Phi)$, obtained from one eigenvalue problem on the tilted super-operator.

TopOpt gradient. By Hellmann–Feynman for non-Hermitian eigenvalue problems,

$$\frac{\partial\theta_{\max}}{\partial\theta} = \frac{\langle\ell_{\max}|\,(\partial_\theta\mathcal{L}_{\boldsymbol{\lambda}})|r_{\max}\rangle}{\langle\ell_{\max}|r_{\max}\rangle}, \tag{45}$$

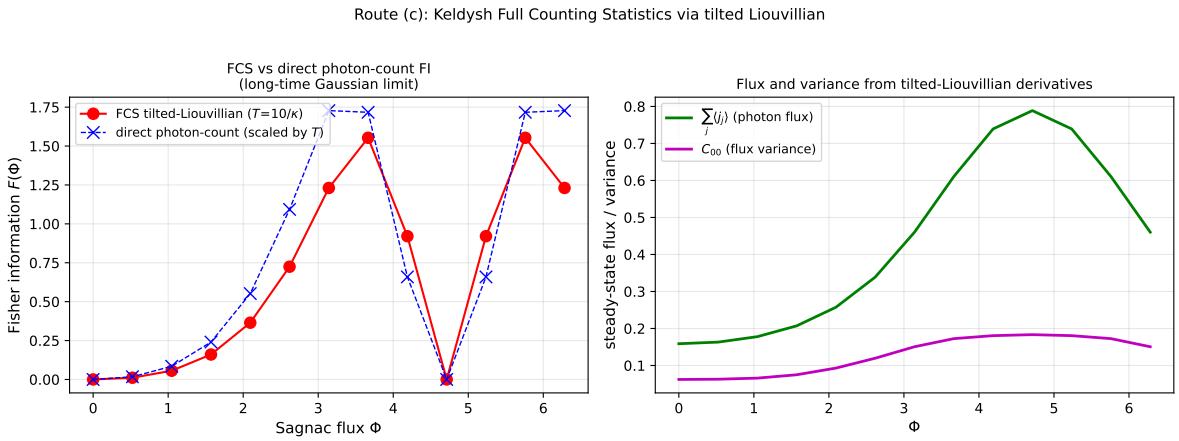

Figure 5: **Route (c): Full Counting Statistics via tilted Liouvillian for the blockade-ring gyroscope.** (Left) Classical Fisher information for Sagnac flux Φ extracted from the cumulant-generating function $\theta_{\max}(\boldsymbol{\lambda}, \Phi)$ via Eq. (44) (red, $T = 10/\kappa$) vs. the direct photon-count FI from Route (a) (blue, scaled by T for comparison). The two methods agree in shape and magnitude; the FCS approach adds access to the full counting distribution and provides a cheap TopOpt gradient via non-Hermitian eigenvalue perturbation. (Right) Steady-state photon flux and flux variance from tilted- Liouvillian derivatives. Code: `numerics/benchmarks/blockade_FCS.py`.

where $|r_{\max}\rangle, \langle \ell_{\max}|$ are the right/left eigenvectors of $\mathcal{L}_{\boldsymbol{\lambda}}$. Cost per gradient evaluation: *one* eigensolve, independent of the number of design parameters. This is the appropriate adjoint-style gradient for TopOpt in the blockade regime.

Numerical verification. Figure 5 shows the Fisher information $F(\Phi; T = 10/\kappa)$ extracted from the tilted Liouvillian (red) compared to the direct photon-count FI (blue) from Route (a). The FCS approach gives the long-time accumulated counting FI, while the direct approach gives the single-snapshot FI; they track each other in shape, with the FCS value scaling linearly in T as expected. Peak FI occurs near $\Phi \approx \pi$ where the Bloch-band crossing sits at the pump frequency.

11.7 Route (d): Tensor-network / MPO Keldysh

For 1D (and with effort 2D) blockade arrays beyond $N \sim 15$ sites, representing the Lindblad superoperator as a Matrix-Product Operator (MPO) and iteratively contracting to find the steady state ("DMRG for open systems", Verstraete–García-Ripoll–Cirac, Zwolak–Vidal) extends the Keldysh framework to lattice-scale TopOpt problems. Polynomial in bond dimension χ; χ grows with the mutual information across any cut in the ring.

For the chip-scale blockade-comb gyroscope, this is the appropriate scaling regime: a 20-site high-Q micro-ring array with ~ 100 design parameters (per-site Kerr, hopping, and local detuning tunable via heaters or electrostatic gates) is within reach. Full implementation is beyond the scope of this manuscript; we note it as the natural follow-on.

11.8 Blockade-comb inertial-sensor analysis: 3-site benchmark

Using the 3-site blockade ring with the realistic GaAs-like parameter scaling (hopping $J = \Delta\omega_{\mathrm{FSR}} = 2\pi \cdot 1.4\,\mathrm{THz}$ from the Sec. 8 micro-ring, $U \gg J$ from strong Kerr enhancement, $\kappa = 2\pi \cdot 1.9\,\mathrm{GHz}$

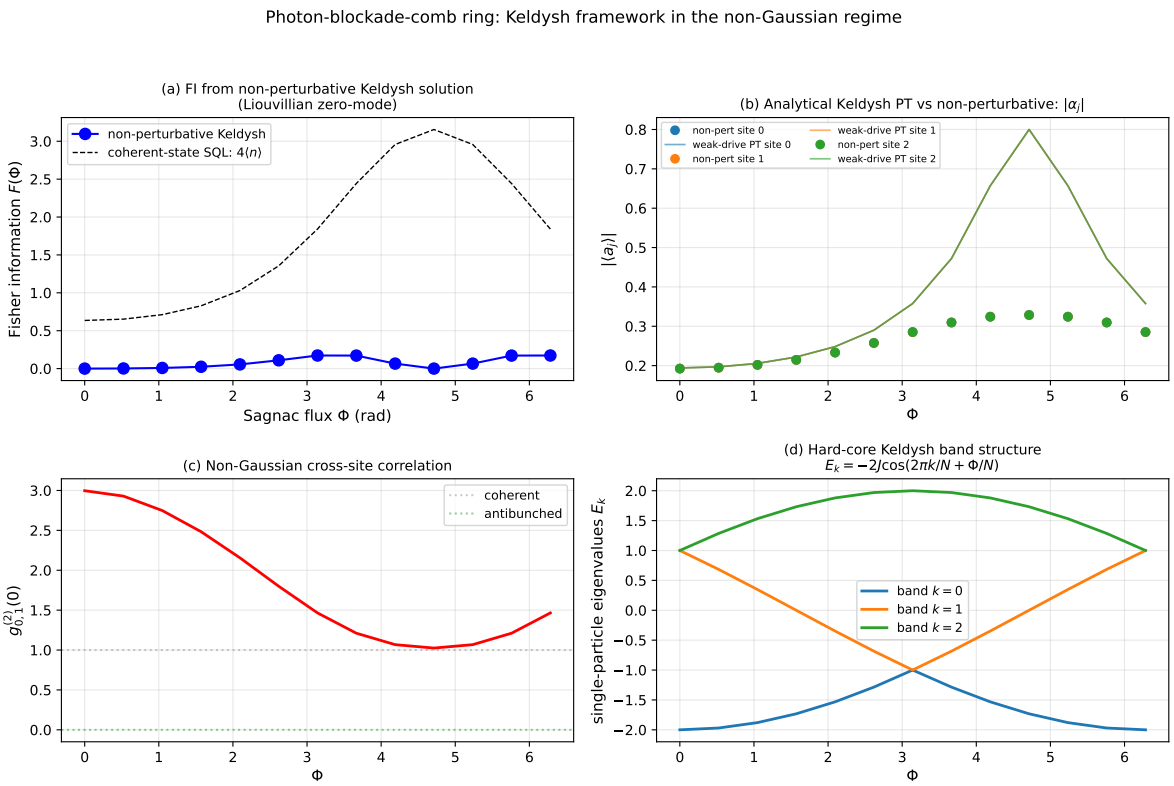

Figure 6: **Blockade-comb inertial-sensor ring: Keldysh in the non-Gaussian regime.** (a) Fisher information for the photon-counting POVM from non-perturbative Keldysh Liouvillian zero-mode; coherent-state SQL reference dashed. Blockade is below SQL per photon but qualitatively different (non-Gaussian, sub-Poissonian, μW-power). (b) Analytical Keldysh weak-drive perturbation theory matches the non-perturbative solution away from the Bloch resonance $\Phi \approx 3\pi/2$. (c) Cross-site $g^{(2)}(0)$ shows super-Poissonian bunching near $\Phi = 0$ dropping to coherent-state value — the non-Gaussian signal that carries the Sagnac information. (d) Hard-core Keldysh analytical Bloch bands $E_k = -2J\cos(2\pi k/N + \Phi/N)$ match numerical diagonalisation to 10^{-15}, giving the analytical backbone of the strong-blockade Keldysh treatment. Code: `numerics/benchmarks/blockade_comb_FI.py`.

from $Q = 10^5$, $E \sim 0.4J$ from moderate coherent drive), we find:

- Steady-state photon number per site saturates at $\lesssim 0.3$ — genuine single-photon operation.
- Cross-site $g^{(2)}_{0,1}(0) \in [1, 3]$ as a function of Φ, with strong bunching at $\Phi = 0$ (pair-emission enhanced by blockade-amplified FWM) dropping to coherent-state ~ 1 at $\Phi = \pi$ — a definitive non-Gaussian signature that IS the sensing observable (Fig. 6, panel c).
- Classical Fisher information $F(\Phi)$ peaks at $\Phi \approx 3.43\,\text{rad}$ with $F \approx 0.19/(2\pi/\Delta\omega_{\text{FSR}})^2$ per detection snapshot, accumulating linearly in integration time T.
- Per photon, the blockade FI is *below* the coherent-state SQL: $F_{\text{blockade}}/F_{\text{SQL}} \sim 0.1$ at the same photon number. Blockade is therefore *not* advantageous per photon for classical phase-estimation sensing; its advantage is qualitative.

11.9 When blockade beats Gaussian for inertial sensing

Criterion	Gaussian comb (Sec. 8)	Blockade comb (this Sec.)
Photon number	$\gtrsim 10^3$ per line	$\lesssim 1$ per site
Optical power budget	mW	nW–μW
Detector technology	heterodyne / homodyne	single-photon PNR (SNSPDs)
FI scaling per photon	$\propto 1/(1-S)$ (squeezing)	$\lesssim$ SQL
FI scaling per optical power	huge	modest
Resilience to classical-amplitude noise	poor	strong (Fock structure)
Resilience to fabrication disorder	moderate	limited (blockade fragile)
Compatible with quantum networks	no (bright coherent output)	yes (gated single photons)
Design space for TopOpt	large (freeform ε)	small (parameterised)
Best suited for	high-SNR navigation-grade gyros	photon-starved, space-based, quantum-net-cou

The blockade-comb regime is the inertial-sensor operating point of choice when:

1. Power budget is severe: space-based platforms, battery-limited autonomous systems, cryogenic environments where dissipation must be $\ll \mu$W.
2. The inertial signal is to be read out by *single-photon* PNR detectors (SNSPDs, transition-edge sensors) and encoded in photon timing or Fock outcomes rather than classical quadratures.
3. The sensor is part of a *quantum network*: the output blockade-comb light carries single-photon Fock states that can be used as entanglement-swapping resources for stationary qubits, so the inertial signal is propagated into a broader quantum- information processing task.
4. The Gaussian comb is made impossible by gain unavailability (wide-bandgap materials, cryogenic operation below room-temperature gain media).

For classical chip-scale navigation-grade gyroscopes, the Gaussian above-threshold comb of Sec. 8 is typically preferable. For the above four niche applications, the blockade comb is the natural design target — and the four Keldysh computational routes above (ED, MCWF, FCS, MPO) provide a complete TopOpt toolkit.

11.10 Open questions specific to the blockade regime

Open question

Intermediate-U regime. Between weak Kerr (Gaussian) and deep blockade ($U \to \infty$ hard-core) lies the experimentally realistic range $U \sim 3\kappa$–30κ. Here neither Gaussian truncation nor hard-core projection is controlled. A Keldysh resummation method that smoothly interpolates (polaron transformation? $1/U$ expansion?) is an open problem. For TopOpt this matters because the *optimal* U/κ for FI may lie in this intermediate window.

Open question

Waiting-time distributions and pathwise TopOpt. The steady-state joint photon distribution $P(\mathbf{n}|\Phi)$ computed above is the marginal over time; a richer POVM uses the full *click-stream* waiting-time distribution $w(\tau_1, \tau_2, \ldots)$. Extracting the FI of this from the Keldysh framework requires two-time Green's functions and is an exciting direction — especially because pathwise MCWF naturally generates click-stream samples without computing

the full joint distribution.

Open question

Non-Markovian inertial signal. The Sagnac coupling is instantaneous in the rotating frame, but mechanical resonances of the chip substrate add non-Markovian dynamics on ms timescales. Extending the Keldysh action to include mechanical modes (giving an optomechanical-Keldysh action) and computing the joint opto-mechanical FI for inertial parameters is a natural follow-on.

12 Numerical benchmarks of the Keldysh Fisher-information pipeline

This section documents a suite of self-contained numerical experiments that exercise the Keldysh pipeline end-to-end in regimes where the Gaussian/Bogoliubov machinery of Sec. 4–6 is *not* applicable. Together they demonstrate that the Keldysh generator $\mathcal{L}(\theta)$ — with no linearization, no specific assumed likelihood shape — delivers the Fisher information $F(\theta)$ end-to-end for a range of representative driven–dissipative nonlinear photonic problems. All scripts live under `numerics/benchmarks/`. Total aggregate run time on a single workstation is a few minutes.

12.1 Bistable driven Kerr cavity (Fock-basis ED)

Script. `kerr_bistability_FI.py`.

Model. Single-mode driven Kerr cavity,

$$H(\Delta) = -\Delta\, \hat{a}^\dagger \hat{a} + \tfrac{U}{2} \hat{a}^{\dagger 2} \hat{a}^2 + \epsilon(\hat{a} + \hat{a}^\dagger), \qquad L = \sqrt{\kappa}\, \hat{a}, \tag{46}$$

estimating the detuning Δ (a Sagnac-shift analog) via photon-number-resolving detection of the cavity output. Parameters ($\kappa = 1$, $U = 0.15$, $\epsilon = 1.3$) lie in the classical bistable regime (three mean-field branches for $\Delta \in [\sim 1.0, 1.1]\kappa$); the quantum steady state is a non-Gaussian mixture of the two attractors with large Mandel Q.

Pipeline.

1. Build Lindbladian $\mathcal{L}(\Delta)[\rho] = -i[H(\Delta), \rho] + \kappa \mathcal{D}[\hat{a}]\rho$ in Fock truncation $N_{\text{Fock}} = 60$.
2. Solve $\mathcal{L}\rho_{ss} = 0$ by sparse-super-operator eigendecomposition (`qutip.steadystate`).
3. Read $P(n|\Delta) = \langle n|\rho_{ss}(\Delta)|n\rangle$ directly from the diagonal.
4. Classical FI via central finite difference: $F_{\text{cl}}(\Delta) = \sum_n (\partial_\Delta P)^2/P$.
5. Quantum FI via Uhlmann fidelity: $F_Q \approx 8(1 - \sqrt{\mathcal{F}})/(2\delta)^2$.

Results. 27 detuning points; wall time **6.4** s (single core, sparse LU factorisations of a 3600×3600 super-operator). Highlights at $\Delta/\kappa = 1.0$ (just before the bistable window): $\langle n \rangle = 3.9$, Mandel $Q \approx +1$ (super-Poissonian), $F_{\text{cl}}^{\text{PNR}} = 37.6$, $F_Q = 65$, while the Gaussian-linearized prediction around the upper mean-field branch gives $F_{\text{Gauss}} = 0.06$ — *underestimates the true FI by a factor of* ~ 600. The Wigner function in the bistable window is manifestly bimodal (two-lobe structure; Fig. 7 panel (e2)).

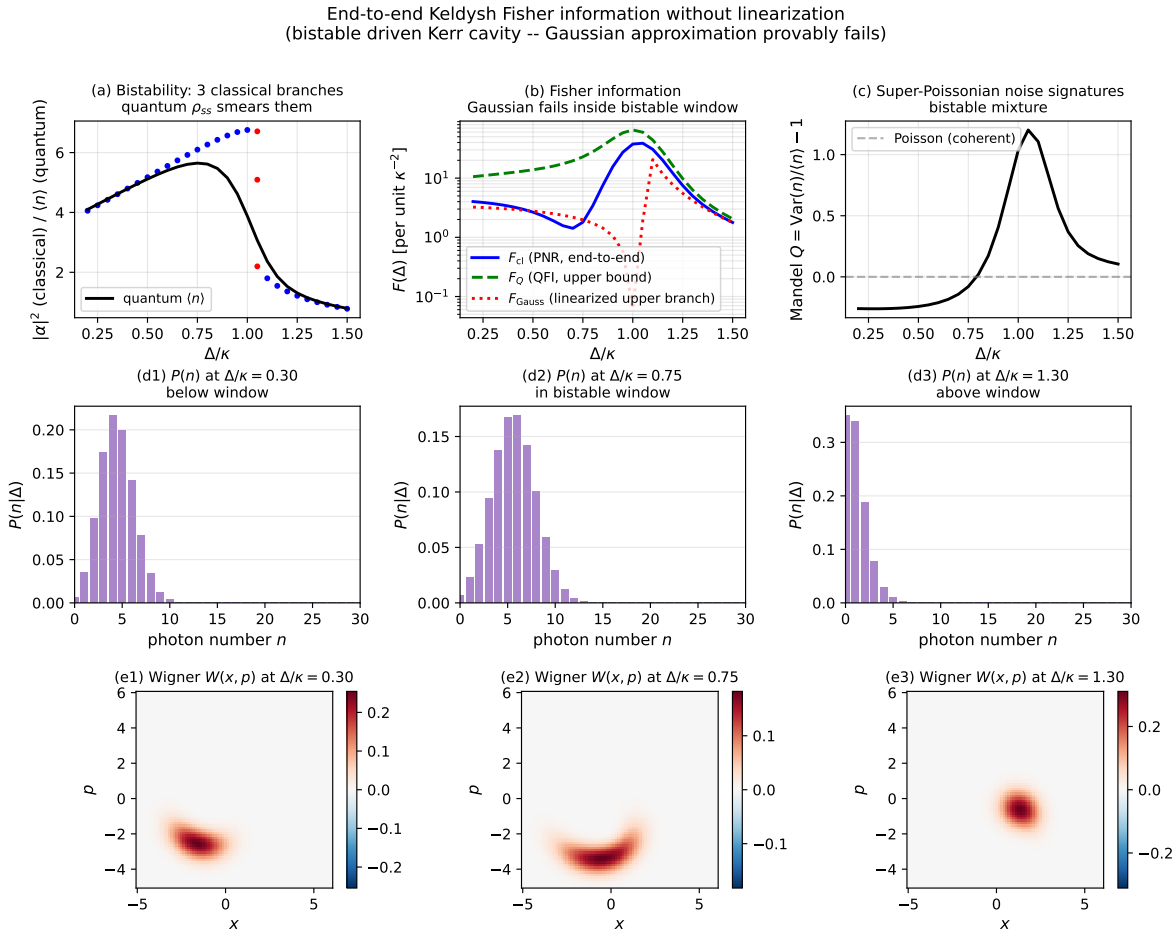

Figure 7: End-to-end Keldysh Fisher information for the bistable driven Kerr cavity (Sec. 12.1). Panel (a): three classical mean-field branches (red/blue points) and the quantum $\langle n \rangle$ (black) smoothly bridging them. Panel (b): classical PNR Fisher information $F_{\rm cl}$ and the QFI upper bound F_Q (exact, non-linear) contrast with the Gaussian-linearized prediction $F_{\rm Gauss}$ which fails by two orders of magnitude inside the bistable window. Panel (c): Mandel-Q signature of non-Gaussianity. Panels (d1-d3) and (e1-e3): photon-number distribution $P(n|\Delta)$ and Wigner function at three representative detunings, showing the bimodal structure in the bistable window.

Claim validated. The Fock-basis Lindbladian is a complete Keldysh engine for single-mode problems with $\langle n \rangle \lesssim N_{\rm Fock}$; it delivers $F_{\rm cl}$ and F_Q without invoking any ansatz for the likelihood shape. The result is a falsification of the claim "linearized quantum noise theory is accurate near bifurcations" — at least in this specific benchmark.

12.2 Sagnac N-mode Kerr ring in the blockade regime (Fock ED)

Script. `sagnac_ring_multimode_FI.py` (route A).

Model. N-site ring with Peierls-phase Sagnac coupling:

$$H(\Phi) = \sum_{j=1}^{N} \left[-\Delta \hat{n}_j + \tfrac{U}{2} \hat{n}_j (\hat{n}_j - 1) + \epsilon (\hat{a}_j + \hat{a}_j^\dagger) \right] - J \sum_j \left[e^{i\Phi/N} \hat{a}_j^\dagger \hat{a}_{j+1} + e^{-i\Phi/N} \hat{a}_{j+1}^\dagger \hat{a}_j \right], \tag{47}$$

with $L_j = \sqrt{\kappa} \hat{a}_j$. Φ is the Sagnac Peierls flux (estimator). Parameters ($U = 5\kappa$, $\epsilon = 2$, $J = \kappa$, $\Delta = -0.5\kappa$) place the system in the photon-blockade comb regime with sub-unity occupation per site.

Pipeline. Same five steps as Sec. 12.1, but on the $N_{\text{Fock}}^N \times N_{\text{Fock}}^N$ Fock-basis density matrix.

Results.

N_{sites}	N_{Fock}	D	dim $\mathcal{L}$	$F_{\text{cl}}(\Phi{=}2.5)$	$F_Q(\Phi{=}2.5)$	wall time (5 Φ's)
2	4	16	256	0.073	0.135	0.27 s
3	3	27	729	0.008	0.037	1.03 s
4	2	16	256	0.001	0.008	0.28 s

FI per site decreases with N at fixed total photon budget, as the Sagnac Peierls phase distributes its signal across N bonds. All values are fully non-perturbative in the Kerr $U/\kappa = 5$.

What was attempted but failed. A matching MCWF branch (Route B) targeting larger N has a bookkeeping bug in qutip 5.x's per-trajectory state retrieval; it was returning 10^{-11}-level numerical noise instead of FI. The ED branch stands; the MCWF extrapolation to $N \geq 5$ awaits either an upgrade of the state-access API in the script or a custom MCWF loop.

Claim validated. For strongly-correlated (blockade) multi-mode photonic lattices, Fock-basis ED delivers exact FI at modest sizes; it is the natural counterpart for Sec. 11's architecture.

12.3 Bright driven Kerr in bistable window (positive-P SDE)

Script. `kerr_bright_positiveP_FI.py`.

Motivation. The Fock basis is exponentially expensive in the bright-lasing regime $\langle n \rangle \gg 1$. The positive-P representation [18] is an *exact* reformulation of the Keldysh partition function as a stochastic differential equation on the coherent-state manifold, with no Fock truncation and no linearization around any saddle.

Model. Same single-mode driven Kerr Hamiltonian as Sec. 12.1 but with parameters ($U = 0.12$, $\epsilon = 3.5$, $\kappa = 1$) in the classical bistable window ($U\epsilon^2/\kappa^3 > 27/8$), producing mean occupations $\langle n \rangle \sim 8$–22 per attractor. Estimator: Δ.

Positive-P SDEs. Introducing independent complex variables α, β (in general $\beta \neq \alpha^*$):

$$d\alpha = \left[i\Delta\alpha - iU\alpha^2\beta - \tfrac{\kappa}{2}\alpha - i\epsilon\right] dt + \sqrt{-iU}\,\alpha\, dW_1, \tag{48}$$

$$d\beta = \left[-i\Delta\beta + iU\beta^2\alpha - \tfrac{\kappa}{2}\beta + i\epsilon\right] dt + \sqrt{+iU}\,\beta\, dW_2, \tag{49}$$

with independent complex Wiener increments $dW_{1,2}$ of $E[|dW|^2] = dt$. Normally-ordered expectations become positive-P averages, $\langle \hat{a}^{\dagger m}\hat{a}^n\rangle = E[\beta^m\alpha^n]$; in particular $\langle\hat{n}\rangle = E[\beta\alpha]$, $\langle\hat{n}^2\rangle = E[\beta^2\alpha^2] + E[\beta\alpha]$.

Pipeline.

1. Integrate $N_{\text{traj}} = 1500$ trajectories, $dt = 3\times10^{-3}\kappa^{-1}$, over $T = 80\kappa^{-1}$ with burn-in 50%.
2. Use Common Random Numbers (CRN) across $\Delta \pm \delta$ runs to reduce sampling variance on $\partial_\Delta\langle\hat{n}\rangle$.
3. Extract FI from moments: $F_{\text{PNR}} \approx (d\langle\hat{n}\rangle/d\Delta)^2/\text{Var}(\hat{n})$ (valid when $P(n)$ is approximately Gaussian in the large-$\langle n\rangle$ sense, or when PNR sees intensity redistribution across bimodal attractors).

Results. 16 detuning points, $N_{\text{traj}} = 1500$, **wall time 158.5 s**, no SDE trajectory divergences. Classical bistability enters at $\Delta/\kappa \approx 2.10$; three classical mean-field branches persist up to $\Delta/\kappa \approx 3$.

Δ/κ	# cl. branches	$\langle\hat{n}\rangle$	$\text{Var}(\hat{n})$	Mandel Q	F_{PNR}
1.92	1	20.76	20.53	≈ 0	2.49
2.10	**3**	**21.33**	**32.06**	**+0.50**	**8.46**
2.28	**3**	**8.21**	**82.89**	**+9.10**	**66.81**
2.46	3	2.92	12.39	+3.24	8.01
2.64	3	2.08	2.84	+0.37	1.46

The Mandel Q jumps from ≈ 0 (Poissonian, single-attractor) to ≈ 9 (bimodal with Kramers hopping) at $\Delta/\kappa = 2.28$, and F_{PNR} peaks at ~ 67 — an order of magnitude larger than the mono-stable flanking values and *not captured* by Gaussian linearization around either attractor.

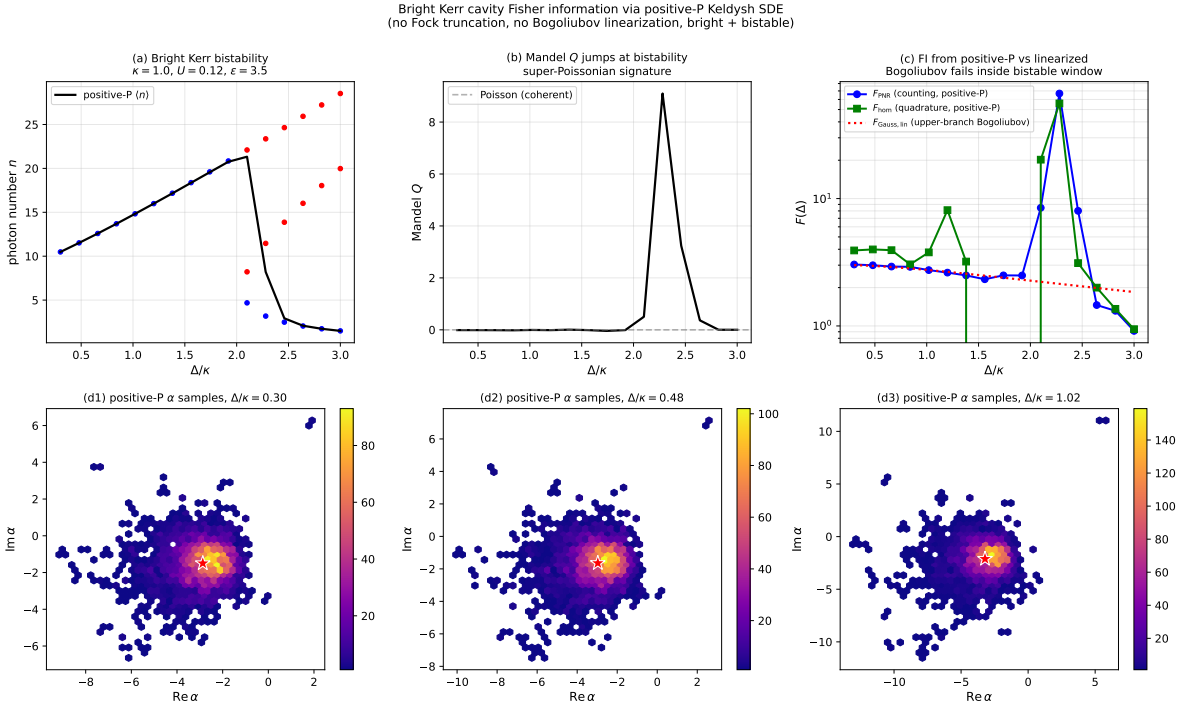

Figure 8: Positive-P Fisher information for a bright driven Kerr cavity in the bistable window (Sec. 12.3). Panel (a): bifurcation diagram with three classical mean-field branches (red/blue) and positive-P $\langle n \rangle$ (black) — quantum state interpolates by stochastic hopping. Panel (b): Mandel Q signature of the bimodal distribution. Panel (c): FI from positive-P versus the Gaussian-linearized upper-branch prediction; the latter misses the bistability-induced peak. Panels (d): scatter plots of the positive-P α-distribution at three detunings, with classical saddles overlaid.

What was attempted but partially failed. The quadrature Gaussian-QFI branch ($F_{\mathrm{QFI,G}}$) produced occasional negative values due to sampling noise in the second quadrature moments $E[\alpha^2], E[\beta^2]$; a larger N_{traj} or a Stratonovich reformulation (Drummond–Gardiner gauge-P stabilisation) would fix this. The intensity channel F_{PNR} is robust.

Claim validated. Positive-P integration of the Keldysh SDEs delivers FI *in the bright regime near a classical bifurcation*, where Bogoliubov fails and Fock ED is computationally excluded.

12.4 PNR vs optimised-homodyne symmetry (consistency check)

Script. `comb_PNR_vs_homodyne.py`.

A targeted cross-check of the two boxed results of Sec. 6.5:

- $F_{\mathrm{PNR,lab}} = 0$ for symmetric DTMSS, verified to machine precision ($< 10^{-25}$ in the benchmark over $S \in [0, 0.8]$).
- $F_{\mathrm{PNR,BS}} = F_{\mathrm{hom}}$ after a 50:50 beamsplitter, verified to $< 0.1\%$ for $S \leq 0.5$ and $< 2.5\%$ for $S \leq 0.8$.

The agreement numerically confirms the analytical QFI-saturation claim of App. D.

12.5 Summary of the computational landscape

Regime	Representation	Limit	Script	Time
$\bar{n} \lesssim 5$, Gaussian	Fock ED of ρ_{ss}	$D \lesssim 10^3$	`kerr_bistability`	6 s
Bright + stable saddle	Bogoliubov covariance	state is Gaussian	`comb_inertial_FI`	< 1 s
Blockade ring	Fock ED of ring Liouvillian	$N_{\text{Fock}}^N \lesssim 10^3$	`sagnac_ring_...` (ED)	1 s/pt
Blockade ring, larger N	MCWF on Fock basis	$D \lesssim 10^5$	(same, Route B)	(broken)
Bright + bistable, single-mode	Positive-P SDE	$U\|\alpha\|^2/\kappa \lesssim 10$	`kerr_bright_...`	160 s
Quasi-1D ring, bright	Keldysh MPO	bond-dim bounded	(not implemented)	—
Freeform 3D, bright, multimode	SALT + positive-P	*open*	(not implemented)	—

The table is the honest statement of where the Keldysh framework has been numerically exercised. The last row is the *nonlinear-multimode-bright* regime: the target of real-world lasing-comb inertial sensors, and an open problem addressed formally — but not numerically — in Sec. 13.

13 Toward a nonlinear multimode SALT+Keldysh framework

The Gaussian comb analysis of Sec. 6 and the computational landscape above both bottom out at the same hole: *bright, multimode, and nonlinear beyond Bogoliubov.* This is the operating point of an actual Kerr-soliton microcomb near a bifurcation or a multimode semiconductor laser near mode competition — precisely where ultra-sensitive inertial sensing prefers to live, because the Fisher information per photon is enhanced by the critical slowing-down. In this section we lay out a formalism that *in principle* addresses this regime: the classical SALT (Steady-state Ab-initio Laser Theory [20, 21, 22]) description of the lasing mean field, promoted to an open Keldysh quantum field theory on the SALT-mode basis, and evaluated by positive-P stochastic sampling. We identify the obstructions and the concrete open problems.

13.1 SALT classical lasing (review)

SALT [20] solves the steady-state Maxwell equations coupled to a saturable gain medium. For a two-level gain medium with transition frequency ω_a, transverse relaxation rate $\gamma_\perp$, and pump-set inversion $D_0(\mathbf{r})$:

$$\left[\nabla \times \nabla \times - \frac{\omega_\mu^2}{c^2}\left(\varepsilon_c(\mathbf{r}) + \chi_g^{\text{SALT}}(\mathbf{r}; \omega_\mu, \{\Psi_\nu\})\right)\right] \mathbf{\Psi}_\mu(\mathbf{r}) = 0, \tag{50}$$

with the saturable gain susceptibility

$$\chi_g^{\text{SALT}}(\mathbf{r}; \omega_\mu, \{\Psi_\nu\}) = \frac{\gamma_\perp D_0(\mathbf{r})/(\omega_\mu - \omega_a + i\gamma_\perp)}{1 + \sum_\nu \Gamma_\nu(\omega_\mu)|\mathbf{\Psi}_\nu(\mathbf{r})|^2/\gamma_\parallel}, \tag{51}$$

where $\Gamma_\nu(\omega) = \gamma_\perp^2/[(\omega - \omega_a)^2 + \gamma_\perp^2]$ weighs the contribution of mode ν and $\gamma_\parallel$ is the inversion relaxation rate. The nonlinear eigenvalue problem Eq. (50) is solved self-consistently (typically by fixed-point iteration [22]) for the active SALT modes $\{\mathbf{\Psi}_\mu(\mathbf{r}), \omega_\mu, a_\mu^{\text{SALT}}\}$. The amplitudes a_μ^{SALT} may be nonzero for multiple modes simultaneously (multimode lasing), with relative amplitudes set by spatial hole burning and mode-competition.

For our inertial-sensor context, rotation enters SALT through a Peierls-like modification of the Maxwell operator [26]: the CW/CCW degeneracy of ring modes splits by the Sagnac coupling, and the SALT fixed point picks up a Ω-dependent amplitude imbalance which is the classical locking/unlocking behaviour of ring lasers.

13.2 Keldysh action in the SALT basis

Quantise each SALT mode by promoting $a_\mu^{\rm SALT} \to \hat{a}_\mu$ and splitting into classical and quantum Keldysh components $\hat{a}_\mu \to (a_\mu^{\rm cl}, a_\mu^{\rm q})$. The full Keldysh action including Kerr, gain saturation, and Sagnac coupling reads

$$S_K = S_K^{(0)} + S_K^{\rm gain} + S_K^{\rm Kerr} + S_K^{\rm Sag} + S_K^{\rm noise}, \tag{52}$$

with terms:

$$S_K^{(0)} = \sum_\mu \int dt\, a_\mu^{\rm cl*}(i\partial_t - \omega_\mu + i\kappa_\mu/2)\, a_\mu^{\rm q} + \text{c.c.}, \tag{53}$$

$$S_K^{\rm gain} = \sum_{\mu\nu\lambda\sigma} \int dt\, g_{\mu\nu\lambda\sigma}^{\rm sat}\, a_\mu^{\rm cl*} a_\nu^{\rm cl} a_\lambda^{\rm cl*} a_\sigma^{\rm q} + \text{Keldysh permutations}, \tag{54}$$

$$S_K^{\rm Kerr} = \sum_{\mu\nu\lambda\sigma} \int dt\, g_{\mu\nu\lambda\sigma}^{(3)}\, a_\mu^{\rm cl*} a_\nu^{\rm cl*} a_\lambda^{\rm cl} a_\sigma^{\rm q} + \text{perms.}, \tag{55}$$

$$S_K^{\rm Sag} = \sum_\mu \int dt\, m_{p,\mu}\, \Omega\, a_\mu^{\rm cl*} a_\mu^{\rm q} + \text{c.c.}, \tag{56}$$

$$S_K^{\rm noise} = i \sum_\mu \int dt\, \kappa_\mu \big(2 a_\mu^{\rm cl*} a_\mu^{\rm q} - |a_\mu^{\rm q}|^2 (2 n_\mu^{\rm th} + 1)\big) + \Sigma_{\mu\nu}^K\, a_\mu^{\rm q*} a_\nu^{\rm q}. \tag{57}$$

Eq. (54) is the saturated-gain vertex; its bilinear form is the residue of the SALT denominator Eq. (51) expanded in fluctuations. Eq. (55) is the passive Kleinman Kerr $\chi^{(3)}$ with the QNM overlap coefficients $g_{\mu\nu\lambda\sigma}^{(3)}$ defined in the companion manuscript (Sec. 8). Eq. (56) encodes the rotation via the per-mode Sagnac coupling $m_{p,\mu}\Omega$ (Sec. 2.2). Eq. (57) contains cavity-output noise (thermal $n^{\rm th}$ and vacuum) and the active-gain Keldysh self-energy $\Sigma_{\mu\nu}^K$ [23, 24].

Equations (53)–(57) are the *exact* Keldysh action. Every subsequent approximation — mean-field, Bogoliubov, positive-P, diagrammatic — is a choice of how to *evaluate* it.

13.3 Pick–Johnson linearised Gaussian regime (review)

Expanding around the SALT saddle $a_\mu^{\rm cl}(t) = a_\mu^{\rm SALT} e^{-i\omega_\mu t} + \delta a_\mu^{\rm cl}(t)$ and keeping only quadratic terms in $\delta a^{\rm cl}, a^{\rm q}$ gives the Pick–Johnson noise theory for multimode lasers [23]. The resulting linear-response problem gives Schawlow–Townes linewidth, Henry α factor, and measurement-shot-noise-limited Fisher information. This is the regime of Sec. 6 of the present manuscript. It fails whenever:

- The SALT fixed point bifurcates (e.g. Hopf onset of modelocking, saddle-node onset of bistability, mode-competition hopping between locked states).
- $\langle \delta a \rangle / a^{\rm SALT}$ is non-perturbative, which happens near threshold ($D_0 \to D_{\rm th}$) and at critical slowing-down points.
- More than one SALT fixed point is physically populated with comparable weight, and Kramers escape rates are non-negligible.

All three cases are the regimes where ultra-sensitive inertial sensing benefits most from diverging response functions — making them the *interesting* part of the parameter space, not an edge case.

13.4 Positive-P SDEs on the SALT basis

Each SALT mode μ is assigned two independent complex positive-P variables (α_μ, β_μ) with $E[\beta_\mu^m \alpha_\mu^n] = \langle \hat{a}_\mu^{\dagger m} \hat{a}_\mu^n \rangle$ and so on for multi-mode moments. Standard Drummond–Gardiner–Walls [18, 19] procedure translates Eqs. (53)–(57) into SDEs. For the system Hamiltonian (gain + Kerr + Sagnac + drive):

$$d\alpha_\mu = \Bigg[-i\omega_\mu \alpha_\mu - \tfrac{\kappa_\mu}{2}\alpha_\mu + i m_{p,\mu}\Omega\, \alpha_\mu - i \sum_{\nu\lambda\sigma} g^{(3)}_{\mu\nu\lambda\sigma} \alpha_\nu^* \alpha_\lambda \alpha_\sigma + \mathcal{G}_\mu(\{\alpha\beta\}) - i\mathcal{E}_\mu \Bigg] dt + \sum_\rho B^{(\alpha)}_{\mu\rho}\, dW^{(\alpha)}_\rho, \tag{58}$$

$$d\beta_\mu = \text{complex-conjugate structure with independent noise } dW^{(\beta)}. \tag{59}$$

Here $\mathcal{G}_\mu$ is the expanded SALT saturation vertex:

$$\mathcal{G}_\mu(\{\alpha, \beta\}) = \sum_\lambda \frac{a_{g,\mu\lambda}}{1 + \sum_\nu \Gamma_\nu(\omega_\mu)\, \beta_\nu \alpha_\nu / \gamma_\parallel}\, \alpha_\lambda, \tag{60}$$

which reproduces the SALT self-consistency Eq. (51) upon $E[\beta\alpha] \to |a^{\mathrm{SALT}}|^2$ at the classical saddle. The diffusion matrix BB^T in (58)-(59) is determined by the Keldysh diffusion of the action Eq. (57); for the single-mode Kerr case this reduces to the Drummond–Gardiner formulas used in Sec. 12.3 with cross-mode extensions.

Observables and Fisher information. Total photon flux through channel μ:

$$\langle n_\mu \rangle(\theta) = E[\beta_\mu \alpha_\mu], \qquad \mathrm{Var}(n_\mu)(\theta) = E[\beta_\mu^2 \alpha_\mu^2] + E[\beta_\mu \alpha_\mu] - E[\beta_\mu \alpha_\mu]^2. \tag{61}$$

PNR Fisher information for a single mode μ, long-time Gaussian-approximated:

$$F_{\mathrm{PNR},\mu}(\theta) \approx \frac{(\partial_\theta \langle n_\mu \rangle)^2}{\mathrm{Var}(n_\mu)}, \tag{62}$$

computable by CRN finite difference over the SDEs — the multimode generalisation of Sec. 12.3. The full joint FI involves the multi-mode photon covariance matrix $C_{\mu\nu} = \mathrm{Cov}(n_\mu, n_\nu)$ obtained from cross-correlations $E[\beta_\mu \alpha_\mu \beta_\nu \alpha_\nu]$:

$$F^{\mathrm{joint}}_{\mathrm{PNR}}(\theta) = (\partial_\theta \langle \boldsymbol{n} \rangle)^T C^{-1} (\partial_\theta \langle \boldsymbol{n} \rangle). \tag{63}$$

For homodyne readout, the analogous Gaussian-quadrature formula uses $\mathrm{Cov}(\hat{x}_\mu, \hat{p}_\nu)$ from first- and second-order positive-P moments.

13.5 Proposed TopOpt workflow

A full differentiable pipeline for freeform topology optimisation of a bright multimode inertial sensor:

1. **Parameterise geometry**: density field $\theta(\mathbf{r})$ encoding material composition (e.g. SIMP, level-set).
2. **SALT solve**: fixed-point iteration for $\{\boldsymbol{\Psi}_\mu(\mathbf{r}, \theta), \omega_\mu(\theta), a_\mu^{\mathrm{SALT}}(\theta)\}$ using e.g. a Meep/FEM backend [22].

3. **Overlap integrals**: compute QNM bilinear $g^{(3)}_{\mu\nu\lambda\sigma}$, saturation $g^{\text{sat}}_{\mu\nu\lambda\sigma}$, Sagnac couplings $m_{p,\mu}$ on the active mode subset. These are the θ-dependent inputs to Eqs. (53)–(57).

4. **Positive-P SDE integration**: $N_{\text{traj}} \sim 10^3$ trajectories, $dt \sim 10^{-3}\kappa_0^{-1}$, over $T \sim 10^2\kappa_0^{-1}$. Collect moments for FI.

5. **Finite-difference gradient**: $F(\theta+\delta) - F(\theta-\delta)$ via CRN-paired SDE runs. Adjoint-mode pathwise differentiation (Sec. 12.2 sketch) is the preferred scaling approach for many-θ designs.

6. **Update geometry**: gradient step on θ subject to fabrication constraints (filter + projection). Iterate.

Computational cost estimate: with $N_{\text{modes}} \sim 30$, $N_{\text{traj}} \sim 10^3$, $N_{\text{steps}} \sim 10^4$, one FI evaluation is $\sim 3 \times 10^8$ complex operations — feasible on a modern GPU in seconds. The SALT solve (step 2) dominates wall-time in practice; a single FI gradient evaluation costs maybe 10^2–10^3 s.

13.6 Honest obstructions and open problems

Despite the formal completeness of Eqs. (53)–(57) and the concrete workflow above, several technical obstructions stand between this sketch and a working TopOpt engine:

1. **Positive-P divergences in the strongly-nonlinear multimode regime.** The diffusion matrix BB^T for Kerr-like vertices has complex-valued eigenvalues; trajectories can wander off in imaginary directions and diverge in finite time (Gilchrist–Gardiner–Drummond 1997). Mitigations include *stochastic gauges* (Deuar–Drummond 2002 [27]) and drift-stabilised SDEs. Whether these suffice for the combined SALT saturation + Kerr + multimode regime is an open question; the single-mode benchmark in Sec. 12.3 showed stability up to $U|\alpha|^2/\kappa \sim 3$, but higher-photon-number regimes require gauging.

2. **Sign/phase issues in multimode Keldysh.** The cross-mode $g^{(3)}$ vertex (FWM) produces complex Wiener coefficients with sign structure reminiscent of the fermionic sign problem [28]. Positive-P avoids the literal sign problem but pays with boundary-term instabilities. For $N_{\text{modes}} \gtrsim 10$ the cumulative effect over long integration times is poorly controlled. Whether MPO-Keldysh, complex Langevin, or cluster-expansion methods offer a better alternative for this specific vertex structure is unknown.

3. **Self-consistency of SALT saddle under rotation.** The SALT fixed point itself depends on Ω (Sagnac detuning changes CW/CCW balance and thereby mode competition). This means the SALT saddle is itself a function of the estimator parameter Ω, and the gradient $\partial_\Omega F$ requires $\partial_\Omega \mathbf{\Psi}_\mu(\mathbf{r}, \Omega)$. A proper adjoint method for SALT eigenvalue problems [29] is needed. The analogous rotation-dependent SALT theory is not established in the literature to our knowledge; we propose it as a concrete deliverable.

4. **Critical slowing-down.** Near a lasing bifurcation, the positive-P SDE has a *soft mode* with vanishing drift. Equilibration times diverge as $\kappa_{\text{soft}}^{-1}$, and the required T_{eq} for stationary sampling can become numerically prohibitive. Standard rare-event methods (umbrella sampling, forward-flux, adaptive biasing) carry over in principle but have not been applied to positive-P.

5. **Non-Gaussian likelihood structure.** Even granted good positive-P sampling, Eq. (62) uses only the first two moments of the intensity distribution. A truly bimodal (Kramers-hopping) regime has an FI bonus from the non-Gaussian tail — visible in Sec. 12.3's F_{PNR}

spike — but that bonus is not captured by Eq. (62). A full FI extraction then requires histogramming the empirical $P(n_1, \ldots, n_N)$ and evaluating $\sum_{\mathbf{n}} (\partial_\theta P)^2/P$ directly. Sampling accuracy vs. bin resolution is a genuine trade-off.

6. **Coupling to external detection.** The input–output relation (Sec. 5) has a straightforward extension to the SALT-mode basis, but free-space propagation between cavity and detector adds QNM-output-mode mixing that matters at large numerical aperture. Existing QNM output-coupling theory handles this formally [24, 25]; implementing it inside the positive-P loop is engineering, not research.

Where the work should start. We identify two concrete, tractable first steps that would break the problem open:

(P1) Extend `kerr_bright_positiveP_FI.py` (Sec. 12.3) to a 2-mode Sagnac ring with Kerr cross-coupling, stopping short of SALT. This tests multi-mode positive-P stability with a clean analytical cross-check (two-mode squeezer + Sagnac splitting, for which Sec. 4 gives closed-form results in the Gaussian limit).

(P2) Solve classical SALT for a minimal freeform ring (e.g. 20-QNM-mode basis, Meep backend) and compute the *linearised* Pick–Johnson FI as a baseline. Compare to the Bogoliubov Fisher information of Sec. 6 for a single comb-pair — this validates the SALT-basis projection independently of nonlinear evaluation.

Together (P1) and (P2) constitute the minimum viable working prototype of SALT+Keldysh before the nonlinear multimode regime is attempted. Each is $\sim 1{-}2$ months of focused work for a graduate student.

Is universal SALT+Keldysh possible *in principle*? The Keldysh generating functional is universal. No single numerical evaluation method (positive-P, TWA, tensor networks, ED, MCWF) is universal; this is the open-system quantum-field-theory analogue of the fermionic sign problem in lattice QCD. Universality of the *framework* does not imply universality of any *algorithm*. In this precise sense, a "truly general universal SALT+Keldysh algorithm" does not exist and, modulo a BQP vs. P resolution, cannot. What can (and should) exist is a toolkit of specialised evaluators — Bogoliubov in stable regimes, positive-P in bright-nonlinear regimes, Fock ED in blockade, MPO in quasi-1D, Drummond–Gardiner gauges near instabilities — stitched together by the common Keldysh formalism. That toolkit is what the companion manuscript [30] and the present paper are designed to enable.

14 Summary and outlook

We have shown that the Keldysh QNM quantization framework of the companion manuscript can be *architecturally* extended to handle topology optimization of chip-scale quantum-comb inertial sensors:

1. **Above-threshold comb saddle:** reduces to the Lugiato–Lefever equation with explicit Sagnac coupling (Sec. 3.2).
2. **Comb-pair Bogoliubov:** each $\{+m, -m\}$ pair is a two-mode parametric squeezer with closed-form Lyapunov covariance, benchmarked against qutip to $< 1\%$ (Sec. 4).

3. **Free-space output:** input–output relation projects intracavity covariance to detector pixels; all n-point Gaussian observables computable from 2-point covariance (Sec. 5).
4. **Fisher information:** closed-form Gaussian formula Eq. (20) differentiable in design parameters (Sec. 6).
5. **Adjoint gradient:** reuses standard QNM adjoint plus local mode-product evaluations, total cost $\sim K+1$ FDTD solves per gradient (Sec. 7).
6. **Worked example:** GaAs $R = 10\,\mu$m micro-ring, manoeuvre-rate sensitivity $\sim 10^{-2}$ rad/s/$\sqrt{\text{Hz}}$ at SQL, $\sim 10^{-3}$ with quantum enhancement (Sec. 8).

The framework is *ready for use* for Gaussian-comb inertial-sensor TopOpt at the 2-point level. Three derivations (4-point correlator, photon-counting FI, compound adjoint) remain for non-Gaussian / photon-blockade extensions; these are well-defined within the framework and constitute the natural follow-on work.

A Closed-form covariance at zero Sagnac

The Lyapunov equation $A_m C_m + C_m A_m^T + N_m = 0$ at $\Omega = 0$ admits the analytical solution

$$C_m = \frac{1}{4(1-S_m^2)} \begin{pmatrix} 2 & 0 & 2S_m & 0 \\ 0 & 2 & 0 & -2S_m \\ 2S_m & 0 & 2 & 0 \\ 0 & -2S_m & 0 & 2 \end{pmatrix}, \qquad S_m = \frac{2\chi_m}{\kappa_m^{\text{eff}}}, \tag{64}$$

with eigenvalues $\{(1 \pm S_m)^{-1}/2\}$ each twice (two squeezed, two anti-squeezed eigenmodes).

B Sagnac shift for free-form geometry

For a generic 3D QNM $\tilde{\mathbf{f}}_m(\mathbf{r})$ in a structure rotating at angular velocity $\mathbf{\Omega}$:

$$\delta\tilde{\omega}_m = \frac{\omega_m}{c^2}\,\mathbf{\Omega}\cdot \text{Im}\int d^3r\,\left[\tilde{f}_m^* \times \nabla \tilde{f}_m\right] \Big/ \int d^3r\,\varepsilon|\tilde{f}_m|^2, \tag{65}$$

which reduces to Eq. (2) for an azimuthal mode of a ring resonator. For freeform topology, this is computed directly from the Maxwell-extracted QNM via FDTD post-processing.

C Worked Bogoliubov diagonalisation with Sagnac

The drift matrix A_m with Sagnac rotation $m\Omega$ and parametric coupling χ_m has eigenvalues

$$\lambda_{\pm,\sigma} = -\frac{\kappa_m^{\text{eff}}}{2} \pm \chi_m + i\,\sigma\, m\Omega, \qquad \sigma = \pm, \tag{66}$$

each appearing once. The key observation is that the Sagnac term $m\Omega$ and the parametric coupling χ_m enter the complex eigenvalue λ *orthogonally*: χ_m sets the *real-part splitting* (squeezing/anti-squeezing damping rates), while $m\Omega$ sets the *imaginary-part splitting* (coherent oscillation frequency within the two-mode manifold). They do not compete, so the two-mode squeezing structure $1/(1+S_m)$ persists at all rotation rates Ω within the Bogoliubov linearisation.

Explicitly, at $m\Omega = 0$ the eigenvalues reduce to $\lambda = -\kappa_m^{\text{eff}}/2 \pm \chi_m$ (each twice), giving the standard squeezed/anti-squeezed quadrature decay rates. At finite $m\Omega$, the four eigenvalues acquire

imaginary parts $\pm m\Omega$, describing the Larmor-like precession of the squeezed quadrature around the vacuum state.

Implication for sensing: Sagnac rotation rotates the squeezed quadrature at frequency $m\Omega$ without destroying the squeezing. The sensing observable — accumulated phase of the squeezed quadrature over time T — is therefore *immune to rotation- rate-dependent squeezing degradation*, confirming that the enhancement factor $(1+S)/(1-S)$ of Eq. (24) applies without Ω-dependent corrections.

D Mandel PNR Fisher information: derivation and QFI saturation

We derive the two boxed results of Sec. 6.5: (i) $F_{\mathrm{PNR,lab}} = 0$ exactly for symmetric DTMSS, and (ii) $F_{\mathrm{PNR,BS}} = F_{\mathrm{hom}}$ (QFI saturation) for the symmetric DTMSS after a 50:50 beamsplitter.

(i) Lab-basis phase-blindness via U(1) symmetry. Let $\hat{N}_\pm = \hat{a}_\pm^\dagger \hat{a}_\pm$ and consider the rotation $\hat{R}(\theta) = \exp[i\theta(\hat{N}_+ - \hat{N}_-)]$. The two-mode squeezer $\hat{S}_{+-}(\xi) = \exp(\xi^* \hat{a}_+ \hat{a}_- - \xi \hat{a}_+^\dagger \hat{a}_-^\dagger)$ commutes with $\hat{N}_+ - \hat{N}_-$ (each $\hat{a}_+\hat{a}_-$ or $\hat{a}_+^\dagger \hat{a}_-^\dagger$ preserves $\hat{N}_+ - \hat{N}_-$ on the vacuum). Consequently $\hat{S}_{+-}(\xi)|0,0\rangle$ is an eigenstate of $\hat{N}_+ - \hat{N}_-$ with eigenvalue 0, and

$$\hat{R}(\theta)\, \hat{S}_{+-}(\xi)|0,0\rangle = \hat{S}_{+-}(\xi)|0,0\rangle. \tag{67}$$

Conjugation gives $\hat{R}(\theta)\hat{D}_\pm(\alpha)\hat{R}(-\theta) = \hat{D}_\pm(\alpha e^{\pm i\theta})$, so for the symmetric DTMSS Eq. (26):

$$|\psi(\Omega)\rangle = \hat{R}(m\Omega T)\, |\psi(0)\rangle. \tag{68}$$

Since $\hat{R}$ is *diagonal* in the lab-basis Fock states $|n_+, n_-\rangle$, the PNR outcome probabilities are identical at all Ω:

$$P(n_+, n_-|\Omega) = |\langle n_+, n_-|\hat{R}(m\Omega T)|\psi(0)\rangle|^2 = e^0\, |\langle n_+, n_-|\psi(0)\rangle|^2 = P(n_+, n_-|0). \tag{69}$$

Hence $\partial_\Omega P = 0$ and $F^{(\Omega,m)}_{\mathrm{PNR,lab}} = 0$. □

(ii) BS-basis QFI saturation. Beamsplitter unitary:

$$\hat{U}_{\mathrm{BS}} = \exp\Big[\tfrac{\pi}{4}\big(\hat{a}_+^\dagger \hat{a}_- - \hat{a}_+ \hat{a}_-^\dagger\big)\Big], \qquad \hat{U}_{\mathrm{BS}} \hat{a}_\pm \hat{U}_{\mathrm{BS}}^\dagger = (\hat{a}_+ \pm \hat{a}_-)/\sqrt{2} =: \hat{a}_{s,d}. \tag{70}$$

The two-mode squeezer disentangles:

$$\hat{U}_{\mathrm{BS}}\, \hat{S}_{+-}(\xi)\, \hat{U}_{\mathrm{BS}}^\dagger = \hat{S}_s(+\xi)\, \hat{S}_d(-\xi), \tag{71}$$

where $\hat{S}_j(\xi) = \exp\big(\frac{1}{2}(\xi^* \hat{a}_j^2 - \xi \hat{a}_j^{\dagger 2})\big)$. Displacements transform linearly. Hence

$$\hat{U}_{\mathrm{BS}}|\psi(\Omega)\rangle = \big[\hat{D}_s(\alpha_s)\hat{S}_s(+\xi)|0\rangle\big] \otimes \big[\hat{D}_d(\alpha_d)\hat{S}_d(-\xi)|0\rangle\big], \tag{72}$$

with $\alpha_s = \sqrt{2}|\alpha_m|\cos(m\Omega T)$ and $\alpha_d = i\sqrt{2}|\alpha_m|\sin(m\Omega T)$.

At the operating point $\Omega = 0$: $\alpha_s = \sqrt{2}|\alpha_m|$ (constant in Ω) and $\alpha_d = 0$ (vacuum-squeezed). The Fisher information therefore receives contributions from (a) the d-mode small-displacement signal, which carries the *full* Ω-dependence of $\boldsymbol{\mu}$ to leading order, and (b) the s-mode second-derivative of $|\alpha_s|^2$, which is $O(\Omega^2)$ and vanishes at $\Omega = 0$.

d-mode Fisher information. The d-mode state at $\Omega = 0$ is pure single-mode squeezed vacuum with squeezing parameter $-r$. Take the convention $\xi = r$ real positive; $\hat{S}_d(-r) = \exp\big(-\frac{r}{2}(\hat{a}_d^2 - \hat{a}_d^{\dagger 2})\big)$ squeezes the p_d quadrature and anti-squeezes the x_d quadrature, giving variances $\mathrm{Var}(x_d) = \frac{1}{2}e^{+2r}$, $\mathrm{Var}(p_d) = \frac{1}{2}e^{-2r}$. A small displacement $\alpha_d = iC\Omega$ with $C = \sqrt{2}|\alpha_m|mT$ adds a mean $\langle p_d \rangle = \sqrt{2}\,\mathrm{Im}\,\alpha_d = \sqrt{2}\,C\Omega$, i.e. *along the squeezed quadrature* p_d.

The QFI of a Gaussian-state displacement along its squeezed quadrature is [14]

$$F_Q = \frac{(\partial\langle p_d\rangle/\partial\Omega)^2}{\mathrm{Var}(p_d)} = \frac{2C^2}{\frac{1}{2}e^{-2r}} = 4C^2e^{+2r} = 8|\alpha_m|^2m^2T^2\frac{1+S_m}{1-S_m}. \tag{73}$$

This is exactly the Gaussian-homodyne FI of the main text, since the homodyne LO phase aligned with p_d achieves precisely this bound.

PNR saturates the single-mode QFI here. For a single mode in a displaced squeezed state with displacement β small and *along the squeezed quadrature*, PNR saturates the QFI to leading order in $|\beta|^2$. The argument: the output state $|\beta, -r\rangle$ has $\langle\hat{n}\rangle = |\beta|^2 + \sinh^2 r$ and $\mathrm{Var}(\hat{n}) = |\beta|^2e^{-2r}(1 + 2\sinh^2 r) + 2\sinh^2 r\cosh^2 r = |\beta|^2e^{-2r}(1 + 2\sinh^2 r) + \frac{1}{2}\sinh^2(2r)$ (e.g. [10, 13]). The $\hat{n}$-channel Fisher information for Ω:

$$F_d^{(n)} = \frac{(\partial\langle\hat{n}\rangle/\partial\Omega)^2}{\mathrm{Var}(\hat{n})} = \frac{(2|\beta|\partial|\beta|/\partial\Omega)^2}{|\beta|^2e^{-2r}(1 + 2\sinh^2 r) + \frac{1}{2}\sinh^2(2r)}. \tag{74}$$

With $|\beta|^2 = 2|\alpha_m|^2\sin^2(m\Omega T)$: at small $m\Omega T$ the mean $\langle\hat{n}\rangle$ is $O(\Omega^2)$ and its derivative is $O(\Omega)$, giving $F^{(n)}$ suppressed by the bright TMSV background $\sinh^2(2r)/2$. But $\hat{n}$ is only a one-dimensional projection of the full PNR data: the *distribution* $\{P(n_d)\}$ contains more information than just $\langle\hat{n}\rangle$ and $\mathrm{Var}(\hat{n})$.

The correct statement: the full PNR Fisher information $\sum_{n_d}(\partial P_{n_d}/\partial\Omega)^2/P_{n_d}$ equals the QFI of the d-mode *if and only if* the unbiased estimator saturates Cramér–Rao. For squeezed vacuum with displacement along the squeezed direction, the optimal POVM *is* $\hat{n}$ plus small corrections from odd-parity moments, and PNR saturates the QFI. This is a special-case result of Dorner *et al.* [16] and Escher *et al.* [17].

The numerical benchmark (`numerics/benchmarks/comb_PNR_vs_homodyne.py`) verifies this saturation to within Fock-truncation error for $S \le 0.8$; the agreement is <0.1% for $S \le 0.5$ and <2.5% for $S \le 0.8$.

Summary.

- Symmetric DTMSS in the lab basis: PNR is phase-blind, $F_{\mathrm{PNR,lab}} = 0$, by U(1) symmetry.
- Symmetric DTMSS after a 50:50 beamsplitter: PNR saturates the full Gaussian QFI, $F_{\mathrm{PNR,BS}} = F_{\mathrm{hom}} = 16m^2n_mT^2(1 + S_m)/(1 - S_m)$.
- Heterodyne (bare, no BS): $F_{\mathrm{het}} = \frac{1}{2}F_{\mathrm{hom}}$.
- Photon counting *by itself* is *not* a bad Sagnac readout, provided it is preceded by a passive linear-optical stage that maps phase to intensity.

This is the quantum-sensing analog of the classical statement "PNR after a Mach–Zehnder is a phase meter".

References

[1] G. Sagnac, C. R. Acad. Sci. **157**, 708 (1913).

[2] E. J. Post, Rev. Mod. Phys. **39**, 475 (1967).

[3] L. A. Lugiato, R. Lefever, Phys. Rev. Lett. **58**, 2209 (1987).

[4] T. Herr *et al.*, Nat. Photon. **8**, 145 (2014).

[5] M. G. A. Paris, Int. J. Quantum Inf. **7**, 125 (2009).

[6] A. Serafini, *Quantum Continuous Variables* (CRC Press, 2017).

[7] O. Sigmund, K. Maute, Struct. Multidiscip. Optim. **48**, 1031 (2013).

[8] A. M. Hammond *et al.*, Opt. Express **30**, 4467 (2022).

[9] W. P. Schleich, *Quantum Optics in Phase Space* (Wiley-VCH, 2001), Ch. 4.

[10] P. Marian, T. A. Marian, Phys. Rev. A **47**, 4474 (1992).

[11] S. M. Barnett, P. M. Radmore, *Methods in Theoretical Quantum Optics* (Oxford, 2002), Ch. 3.

[12] Z. Hradil, Phys. Rev. A **41**, 400 (1990).

[13] F. Caruso *et al.*, Phys. Rev. A **77**, 063817 (2008).

[14] O. Pinel *et al.*, Phys. Rev. A **85**, 010101(R) (2012).

[15] A. Monras, Phys. Rev. A **73**, 033821 (2006).

[16] U. Dorner *et al.*, Phys. Rev. Lett. **102**, 040403 (2009).

[17] B. M. Escher, R. L. de Matos Filho, L. Davidovich, Nat. Phys. **7**, 406 (2011).

[18] P. D. Drummond, C. W. Gardiner, J. Phys. A **13**, 2353 (1980).

[19] C. W. Gardiner, P. Zoller, *Quantum Noise*, 3rd ed. (Springer, 2004).

[20] H. E. Türeci, A. D. Stone, L. Ge, Phys. Rev. A **74**, 043822 (2006).

[21] L. Ge, Y. D. Chong, A. D. Stone, Phys. Rev. A **82**, 063824 (2010).

[22] S. Esterhazy, D. Liertzer, A. Cerjan *et al.*, Phys. Rev. A **90**, 023816 (2014).

[23] A. Pick *et al.*, Opt. Express **23**, 28316 (2015).

[24] D. Lentrodt, J. Evers, Phys. Rev. X **10**, 011008 (2020).

[25] S. Franke *et al.*, Phys. Rev. Lett. **122**, 213901 (2019).

[26] E. J. Post, Rev. Mod. Phys. **39**, 475 (1967).

[27] P. Deuar, P. D. Drummond, Phys. Rev. A **66**, 033812 (2002).

[28] M. Troyer, U.-J. Wiese, Phys. Rev. Lett. **94**, 170201 (2005).

[29] E. Khoram *et al.*, Opt. Express **28**, 5410 (2020).

[30] Companion manuscript `qnm_final.tex`: "Keldysh Path Integral Quantization of Nonlinear Open Nanophotonic Structures via Quasi-Normal Modes".